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Lee et al.

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(54) **FLEXIBLE MERGE SCHEME FOR
SOURCE/DRAIN EPITAXY REGIONS**

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H10D 62/13 (2025.01)
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(52) **U.S. Cl.**
CPC **H10B 10/12** (2023.02); **H01L 21/0273**
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CPC H01L 27/1104; H01L 27/1116; H01L
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(56) **References Cited**

U.S. PATENT DOCUMENTS

8,836,016 B2 9/2014 Wu et al.
8,841,701 B2 9/2014 Lin et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 103681347 A 3/2014
CN 105977159 A 9/2016
(Continued)

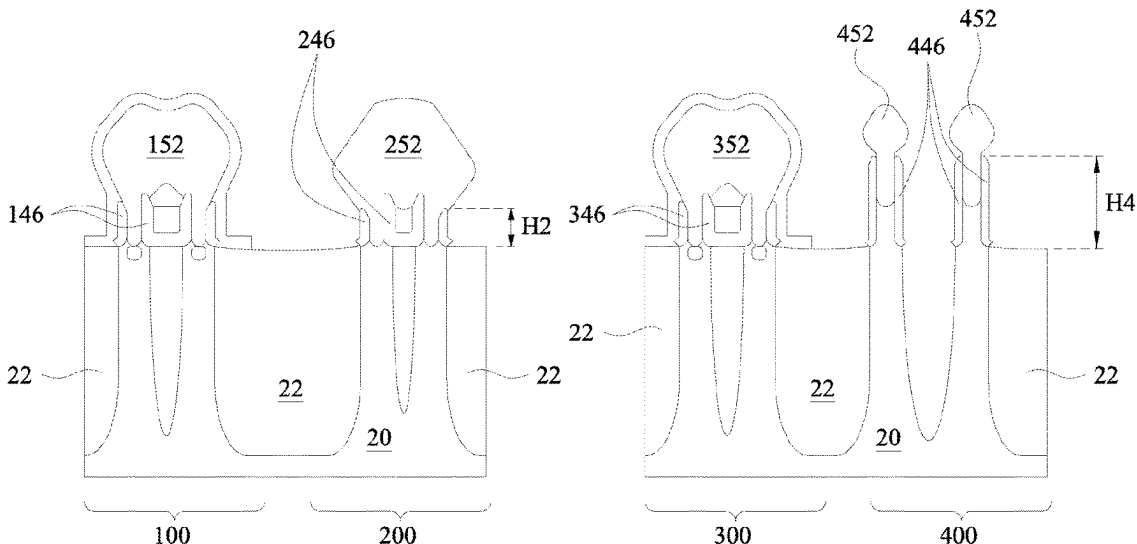
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(57) **ABSTRACT**

A method includes etching a first semiconductor fin and a second semiconductor fin to form first recesses. The first and the second semiconductor fins have a first distance. A third semiconductor fin and a fourth semiconductor fin are etched to form second recesses. The third and the fourth semiconductor fins have a second distance equal to or smaller than the first distance. An epitaxy is performed to simultaneously grow first epitaxy semiconductor regions from the first recesses and second epitaxy semiconductor regions from the second recesses. The first epitaxy semiconductor regions are merged with each other, and the second epitaxy semiconductor regions are separated from each other.

20 Claims, 33 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/669,736, filed on Oct. 31, 2019, now Pat. No. 10,985,167, which is a continuation of application No. 16/206,183, filed on Nov. 30, 2018, now Pat. No. 10,529,725, which is a division of application No. 15/492,142, filed on Apr. 20, 2017, now Pat. No. 10,483,266.

H10D 30/019; H10D 30/0191; H10D 30/0193; H10D 30/0194; H10D 30/0195; H10D 30/0196; H10D 30/0197; H10D 30/0198

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

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H01L 21/311 (2006.01)
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H10D 62/822 (2025.01)
H10D 64/01 (2025.01)
H10D 84/01 (2025.01)
H10D 84/03 (2025.01)
H10D 84/85 (2025.01)

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8,847,293	B2	9/2014	Lee et al.	
8,853,025	B2	10/2014	Zhang et al.	
8,962,400	B2	2/2015	Tsai et al.	
9,093,514	B2	7/2015	Tsai et al.	
9,166,010	B2	10/2015	Kelly et al.	
9,236,267	B2	1/2016	De et al.	
9,245,805	B2	1/2016	Yeh et al.	
9,287,382	B1	3/2016	Lee et al.	
9,520,482	B1	12/2016	Chang et al.	
9,548,362	B2	1/2017	Ching et al.	
9,576,814	B2	2/2017	Wu et al.	
9,640,533	B2	5/2017	Lim et al.	
9,899,268	B2	2/2018	Wei et al.	
10,026,662	B2	7/2018	Sung et al.	
10,483,266	B2 *	11/2019	Lee	H01L 21/31144
10,510,753	B2	12/2019	Lee et al.	
10,529,725	B2 *	1/2020	Lee	H01L 21/823418
10,985,167	B2 *	4/2021	Lee	H01L 21/823821
11,217,585	B2 *	1/2022	Lin	H01L 27/0207
11,581,226	B2 *	2/2023	Lin	H01L 21/823821
11,856,743	B2 *	12/2023	Lee	H01L 29/6656
2014/0065782	A1	3/2014	Lu et al.	
2014/0203338	A1 *	7/2014	Kelly	H10D 30/62 257/288 H10D 84/0193 257/401
2016/0064387	A1 *	3/2016	Jeong	
2016/0190017	A1	6/2016	Lee et al.	
2016/0268171	A1 *	9/2016	Wei	H10D 86/011
2016/0358925	A1	12/2016	Bai et al.	
2017/0077222	A1	3/2017	Lee et al.	
2017/0133286	A1 *	5/2017	Sung	H01L 21/823821
2017/0207126	A1 *	7/2017	Ching	H01L 21/31111
2019/0157155	A1	5/2019	Keng et al.	

FOREIGN PATENT DOCUMENTS

CN	106033757	A	10/2016
CN	106560925	A	4/2017
CN	106684149	A	5/2017
KR	20160137962	A	12/2016
TW	201409581	A	3/2014
TW	201701464	A	1/2017
WO	2015147842	A1	10/2015

* cited by examiner

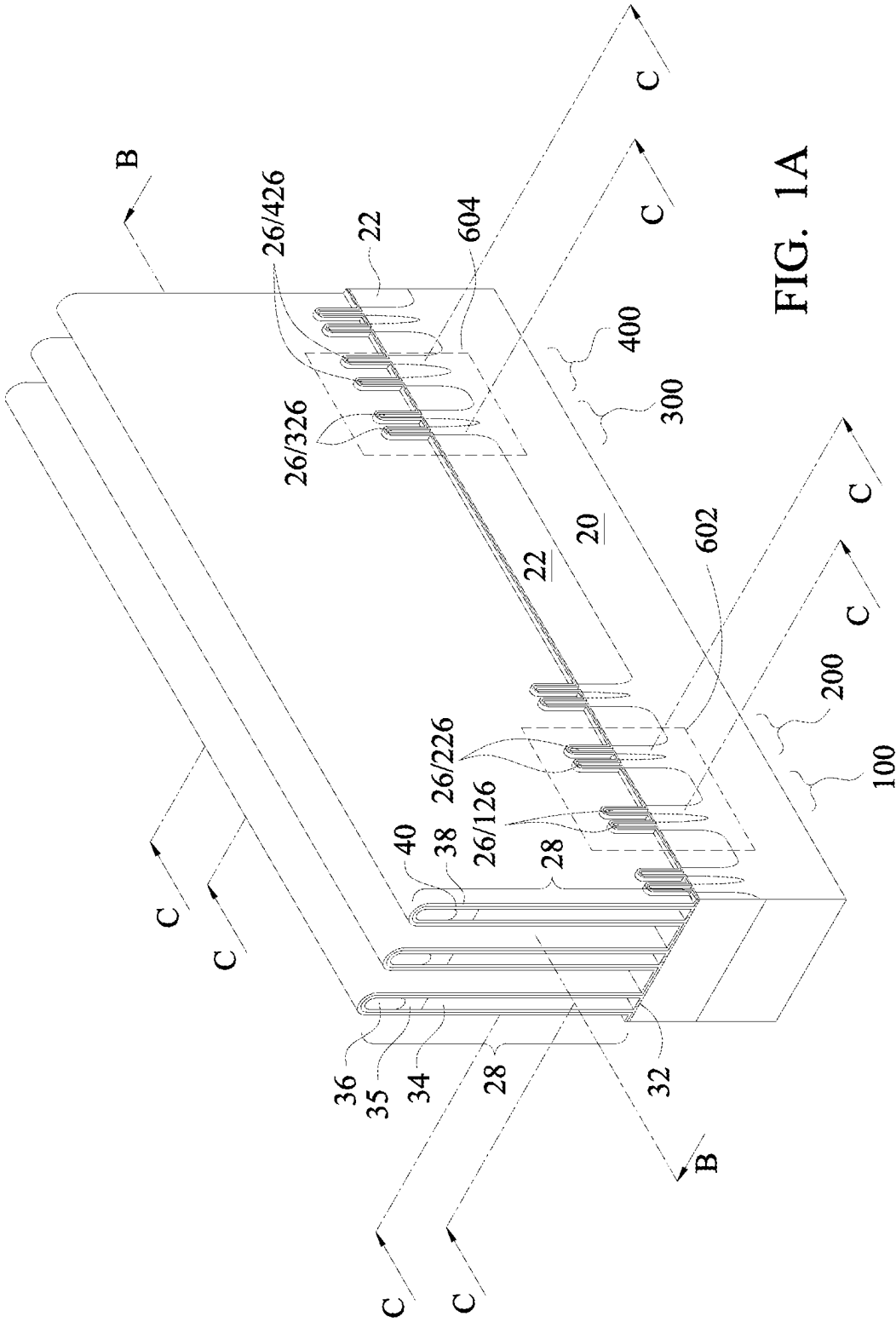


FIG. 1A

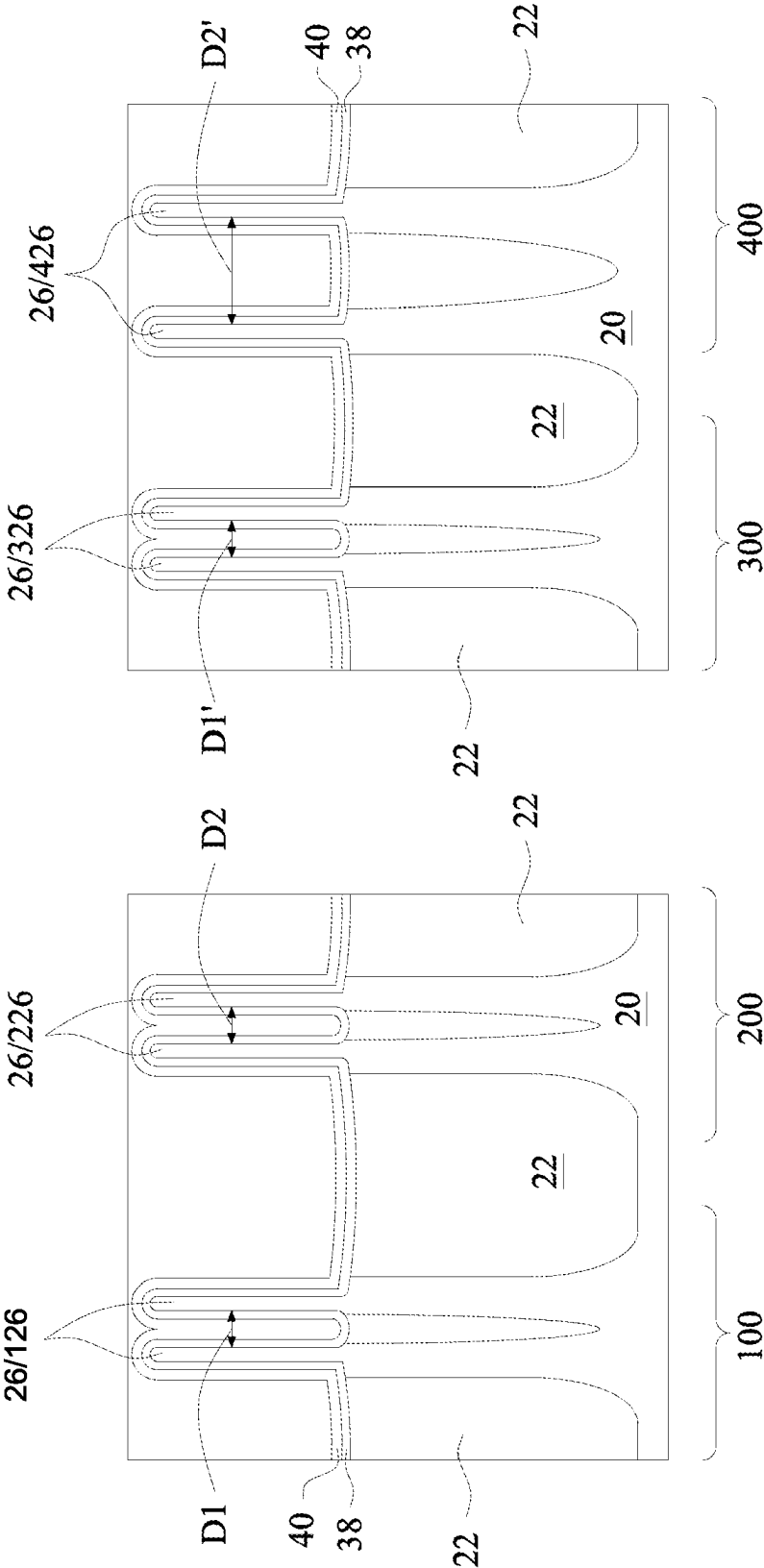


FIG. 1B

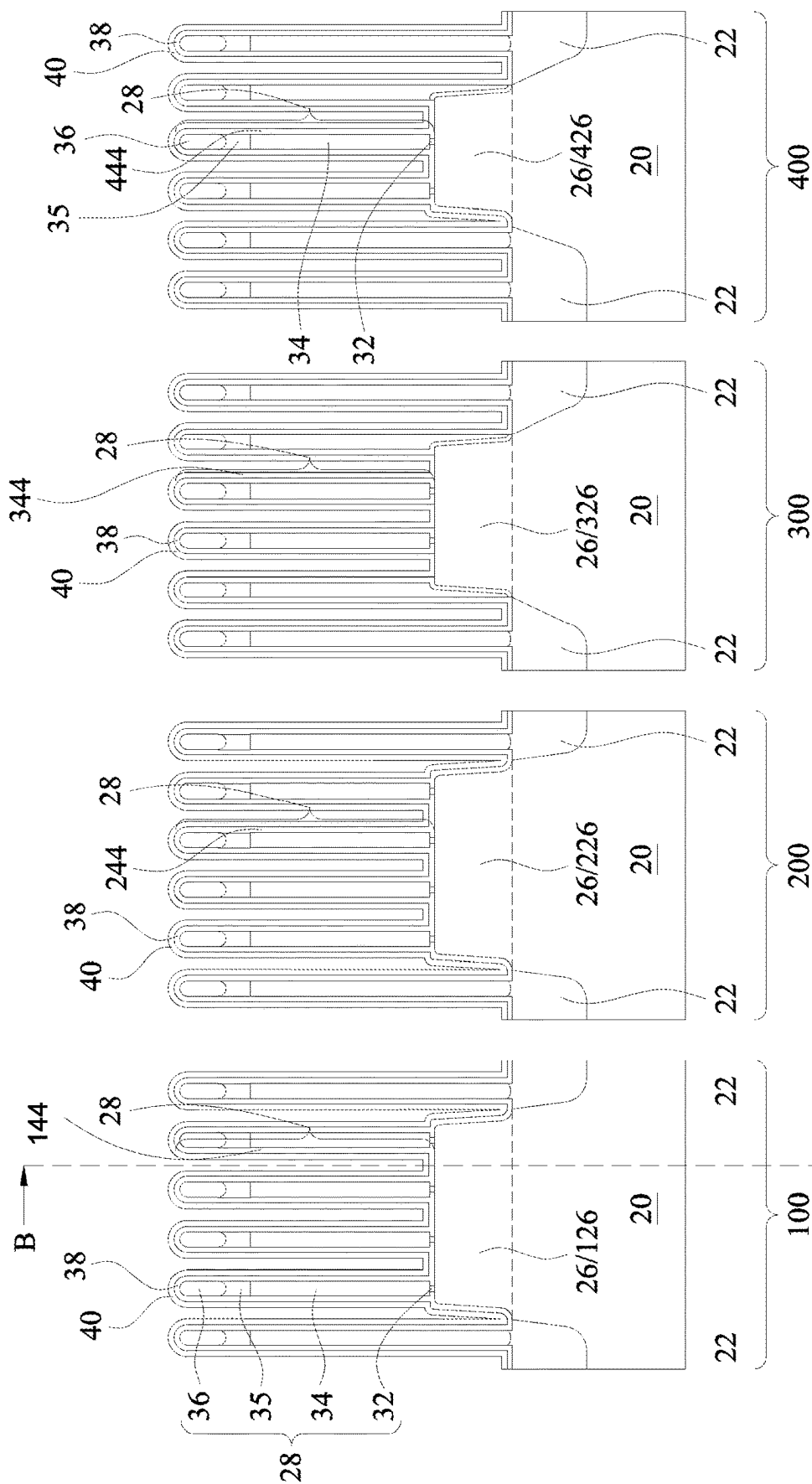


FIG. 1C

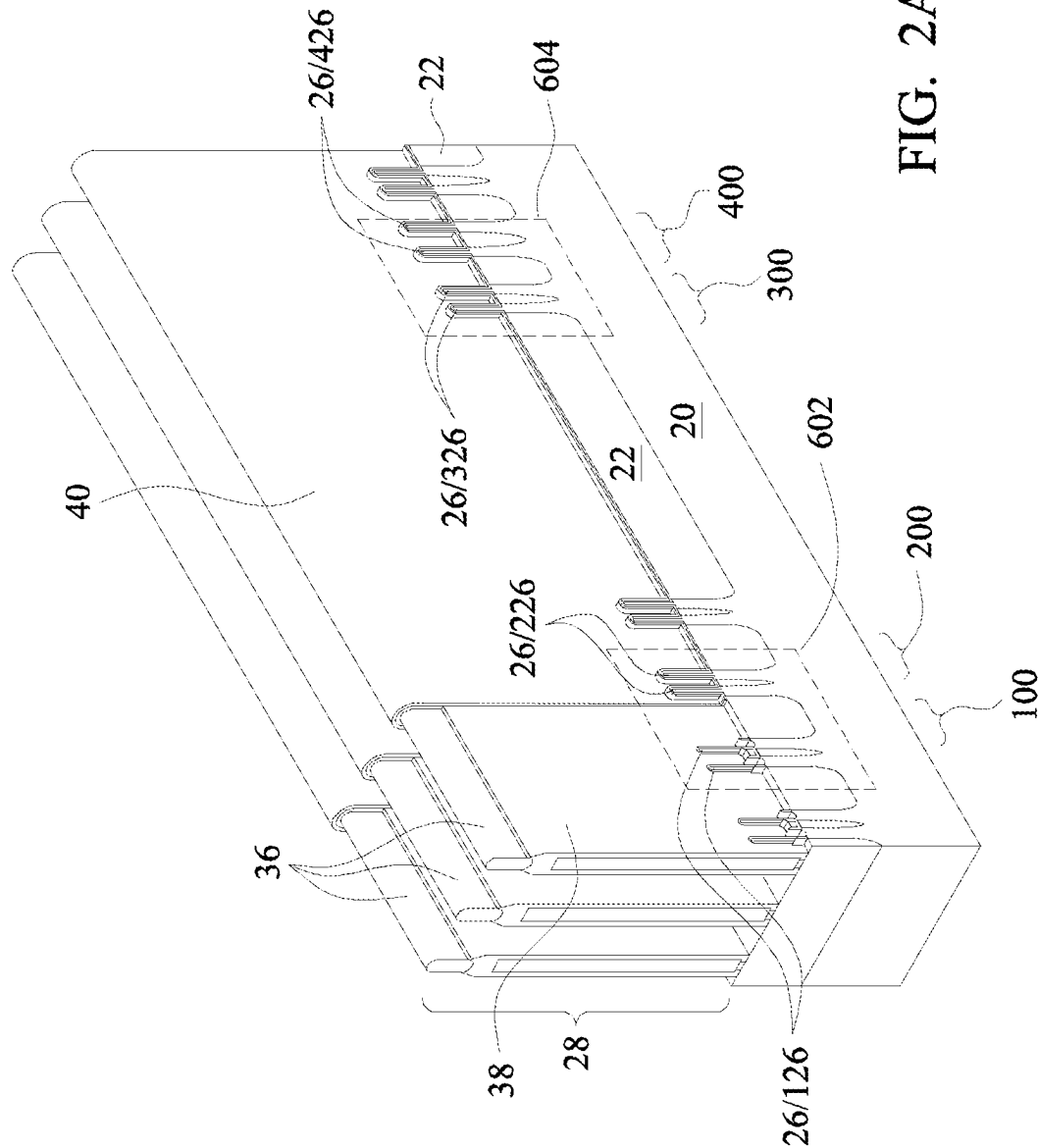


FIG. 2A

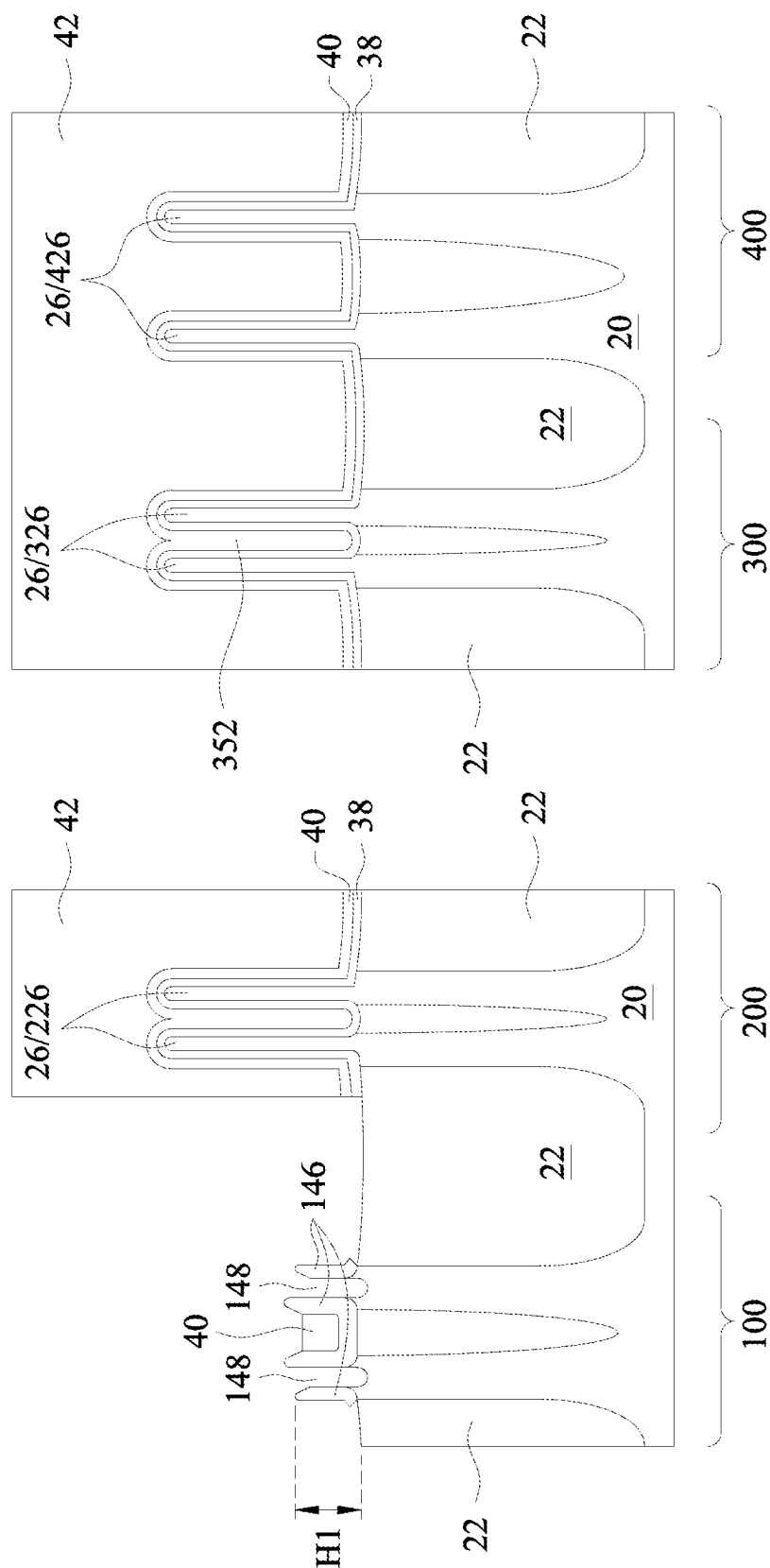


FIG. 2B

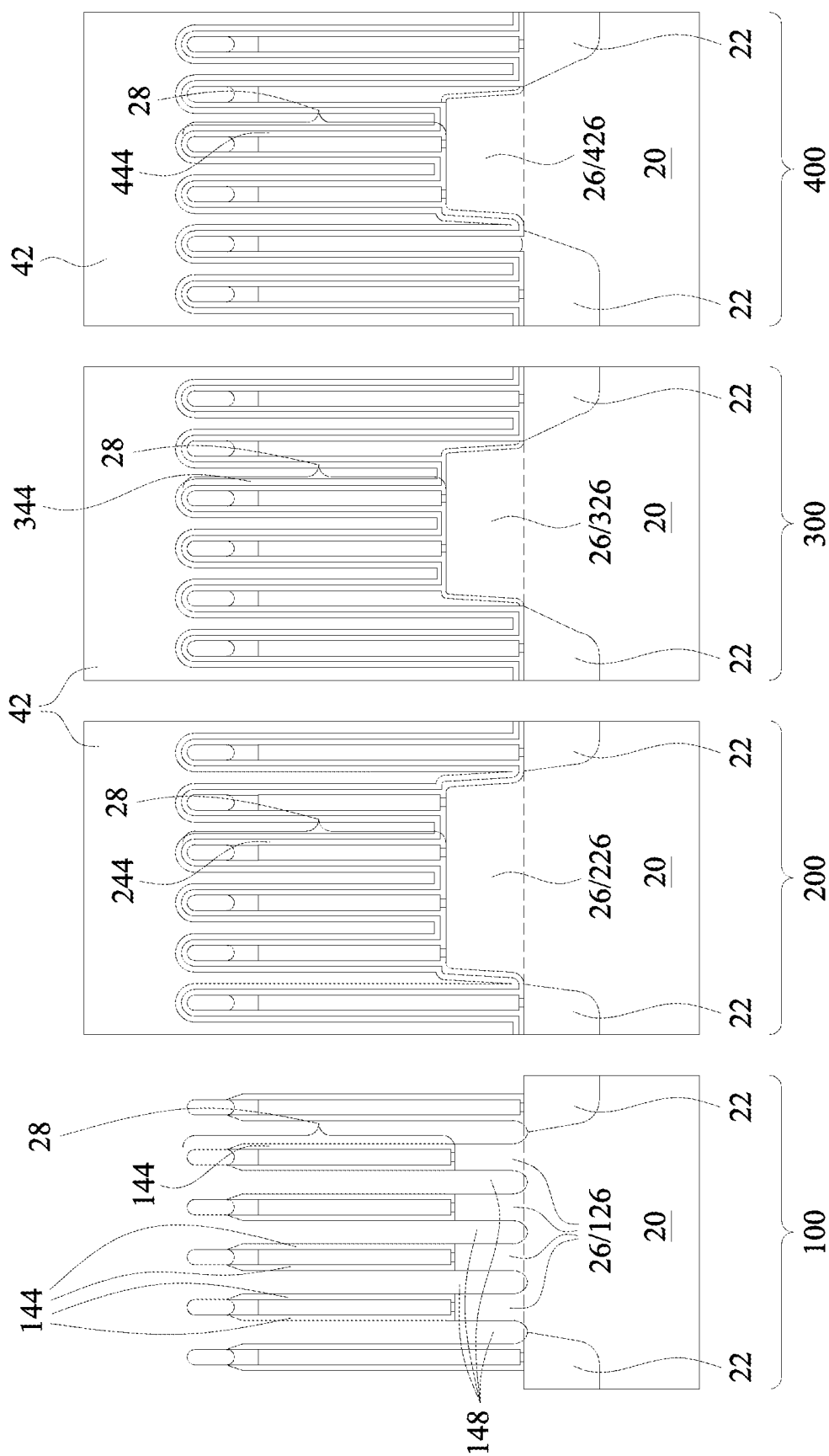


FIG. 2C

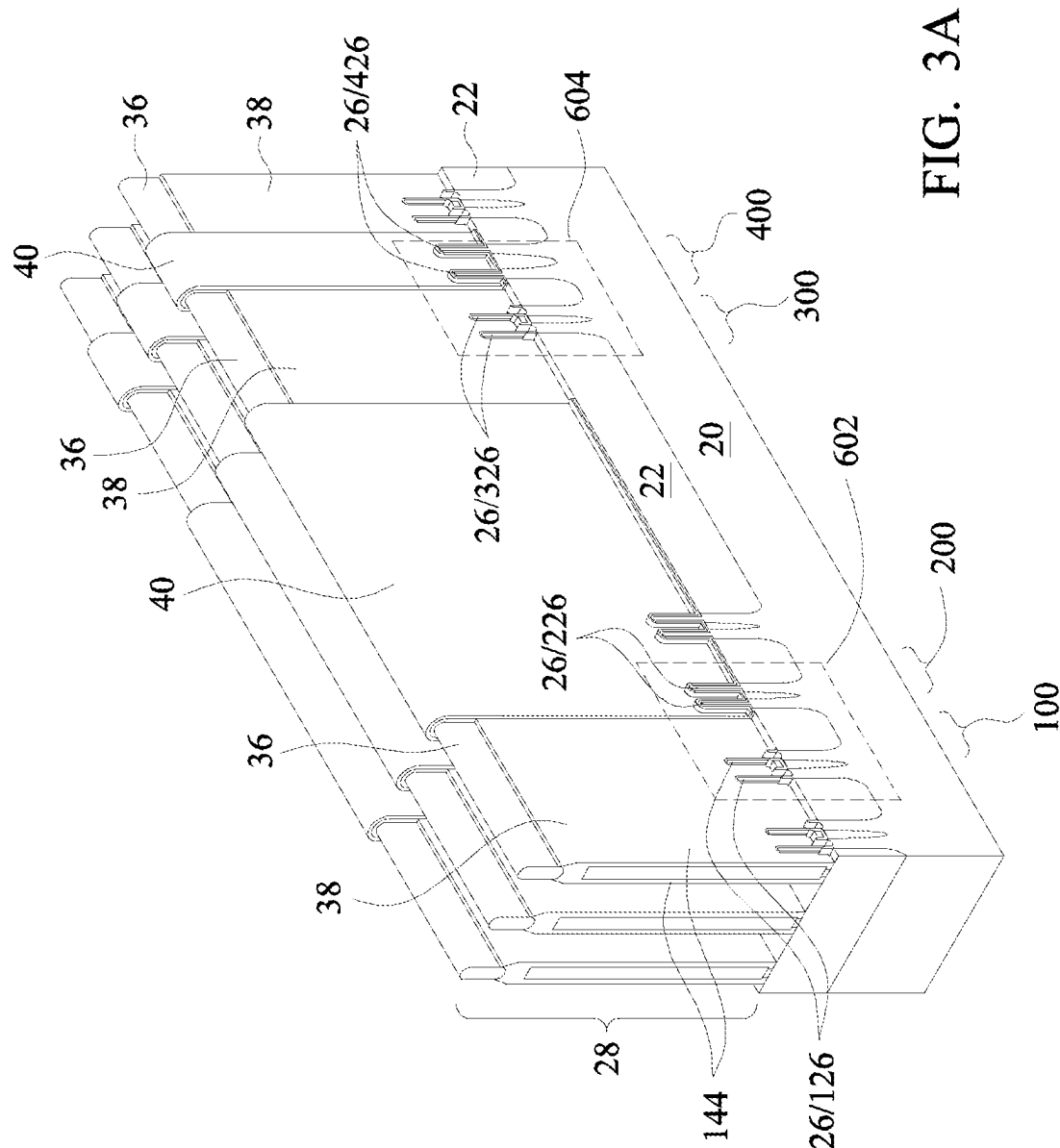


FIG. 3A

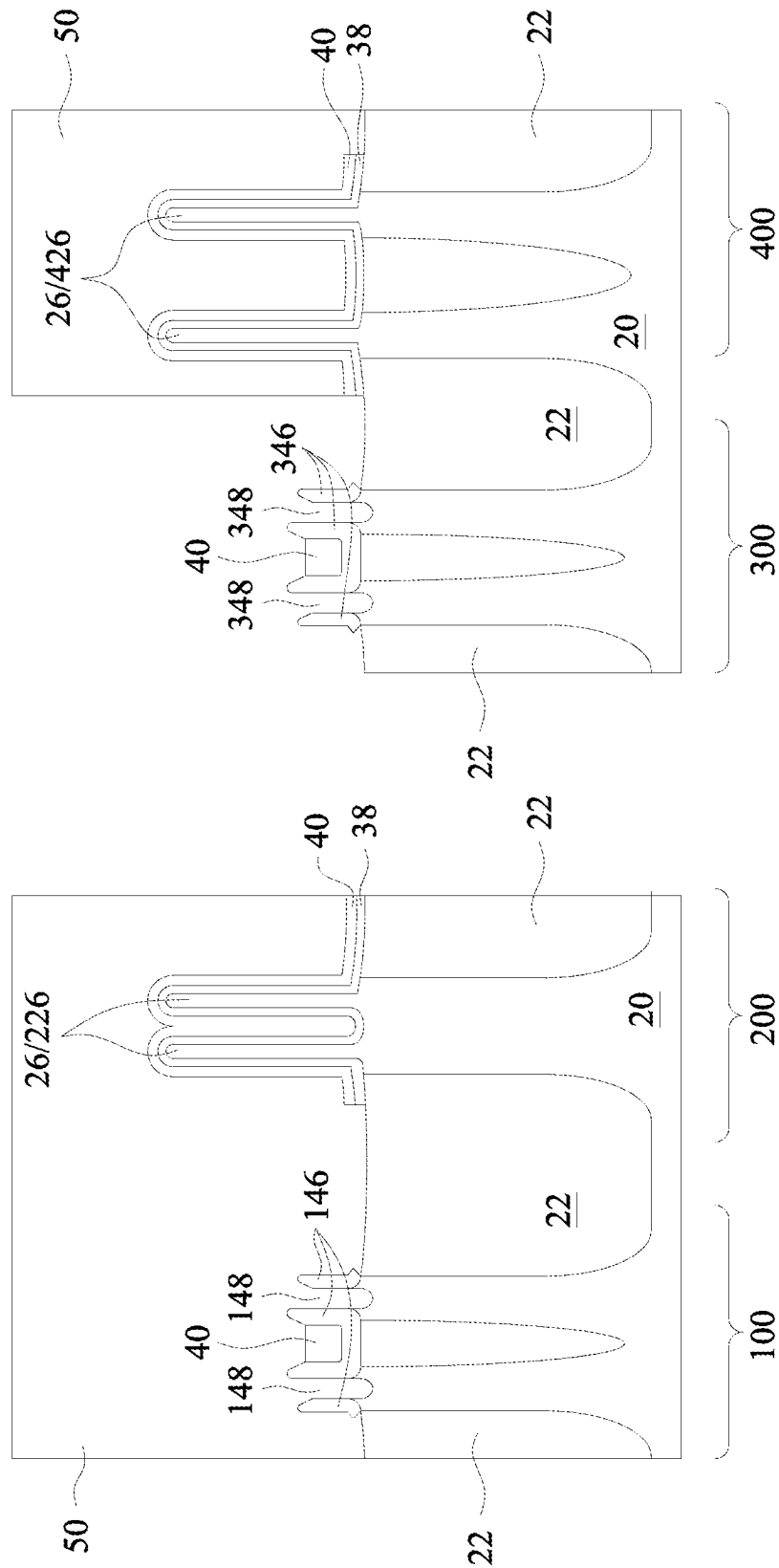


FIG. 3B

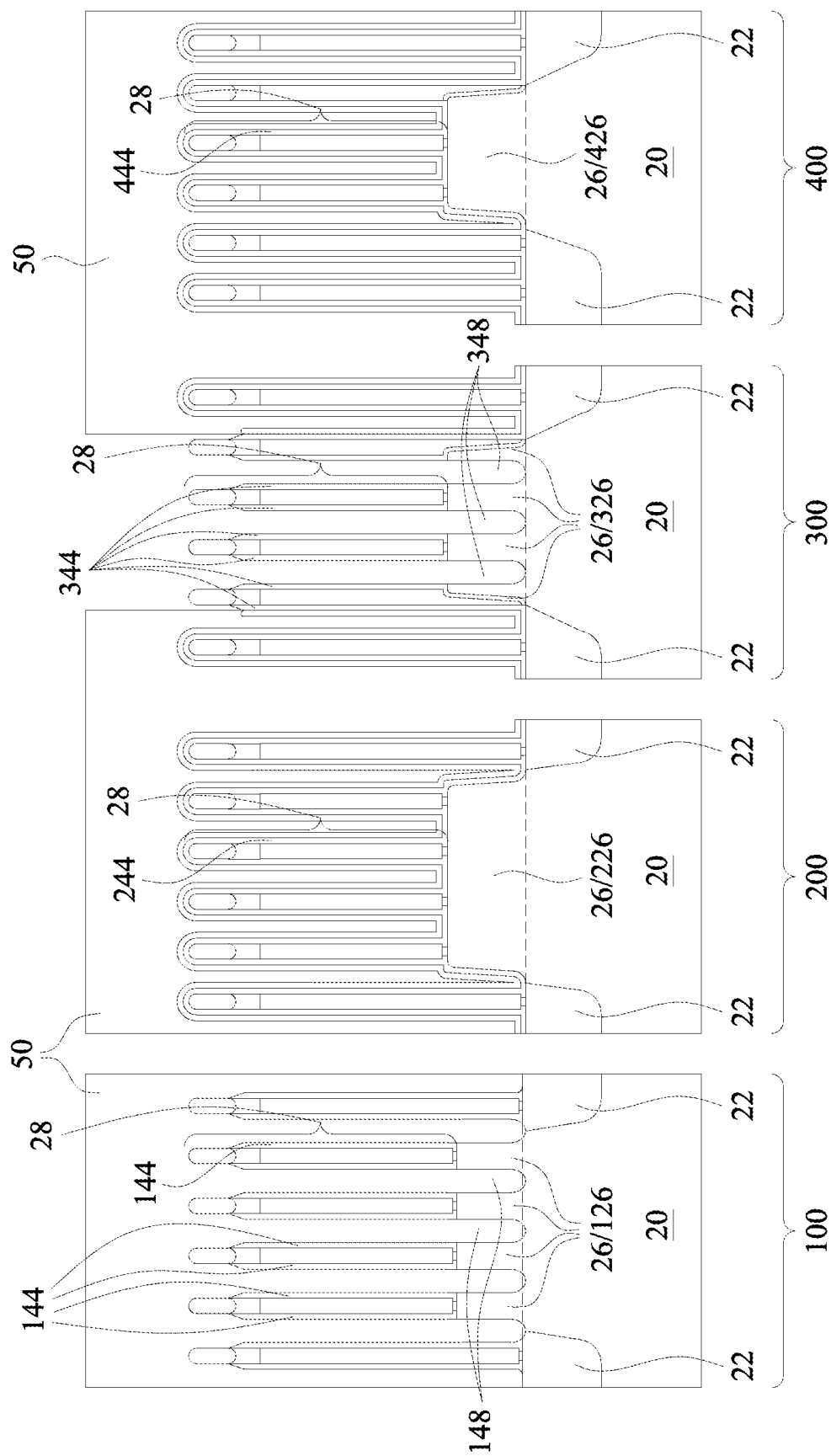
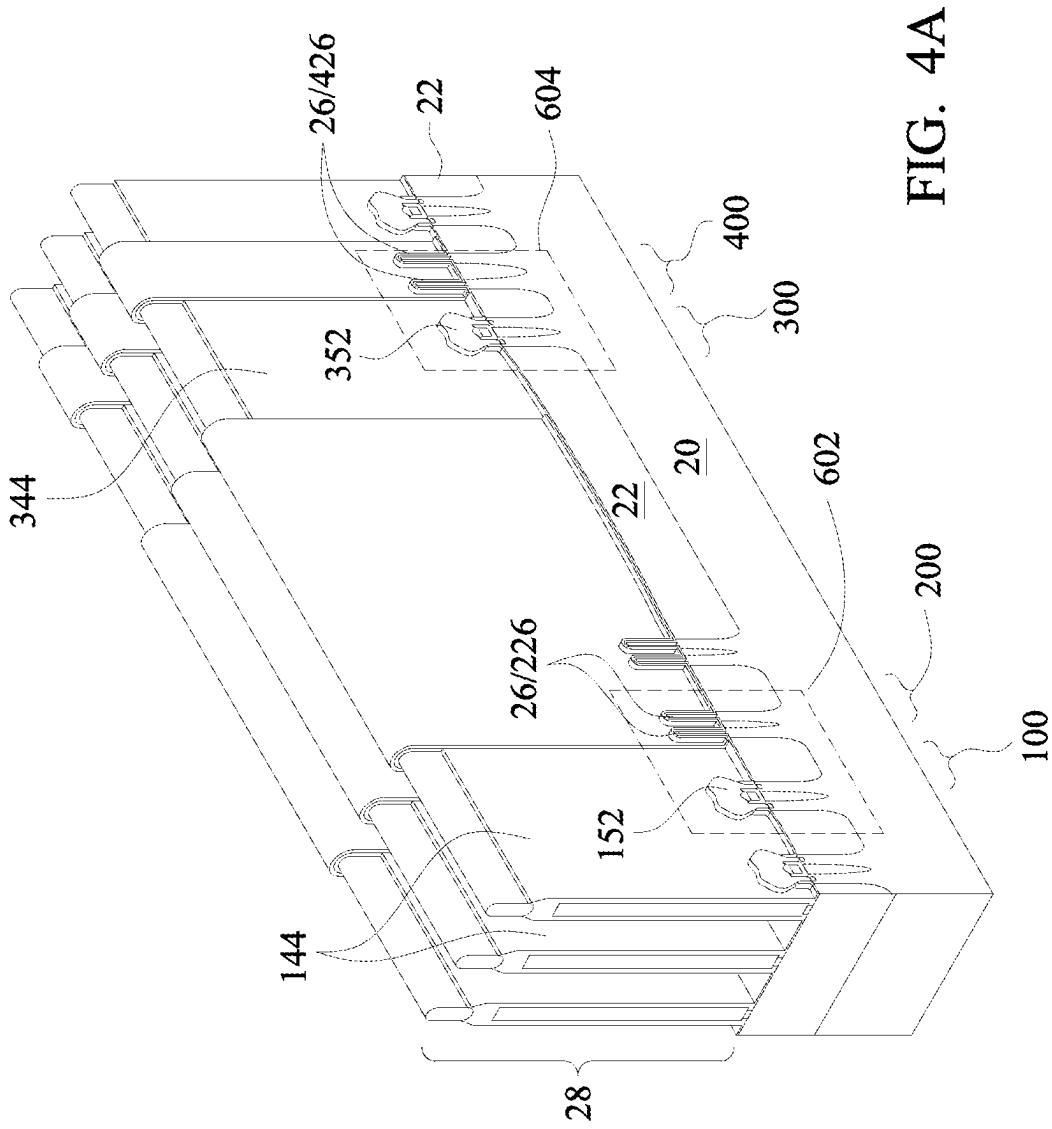


FIG. 3C



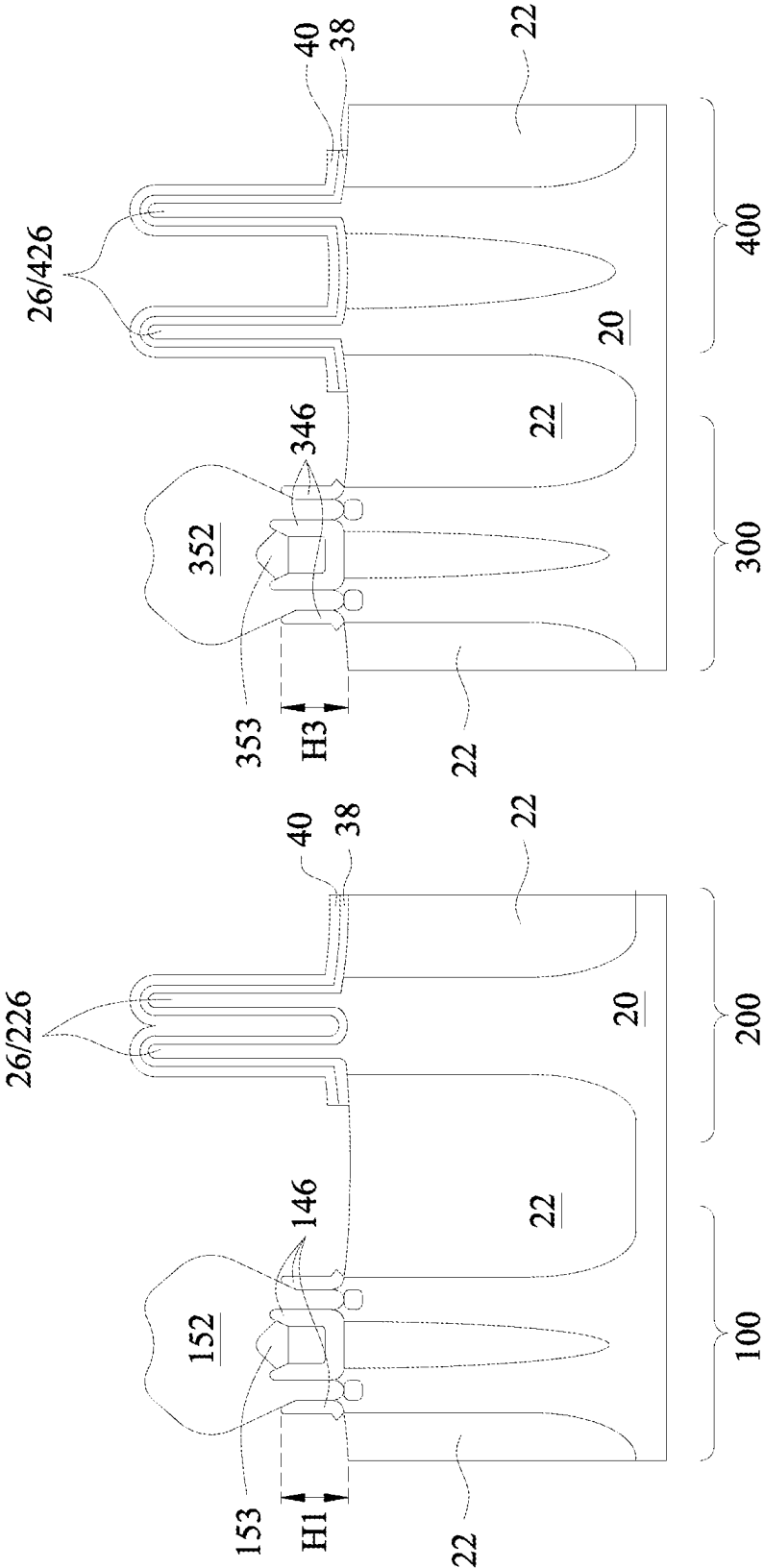


FIG. 4B

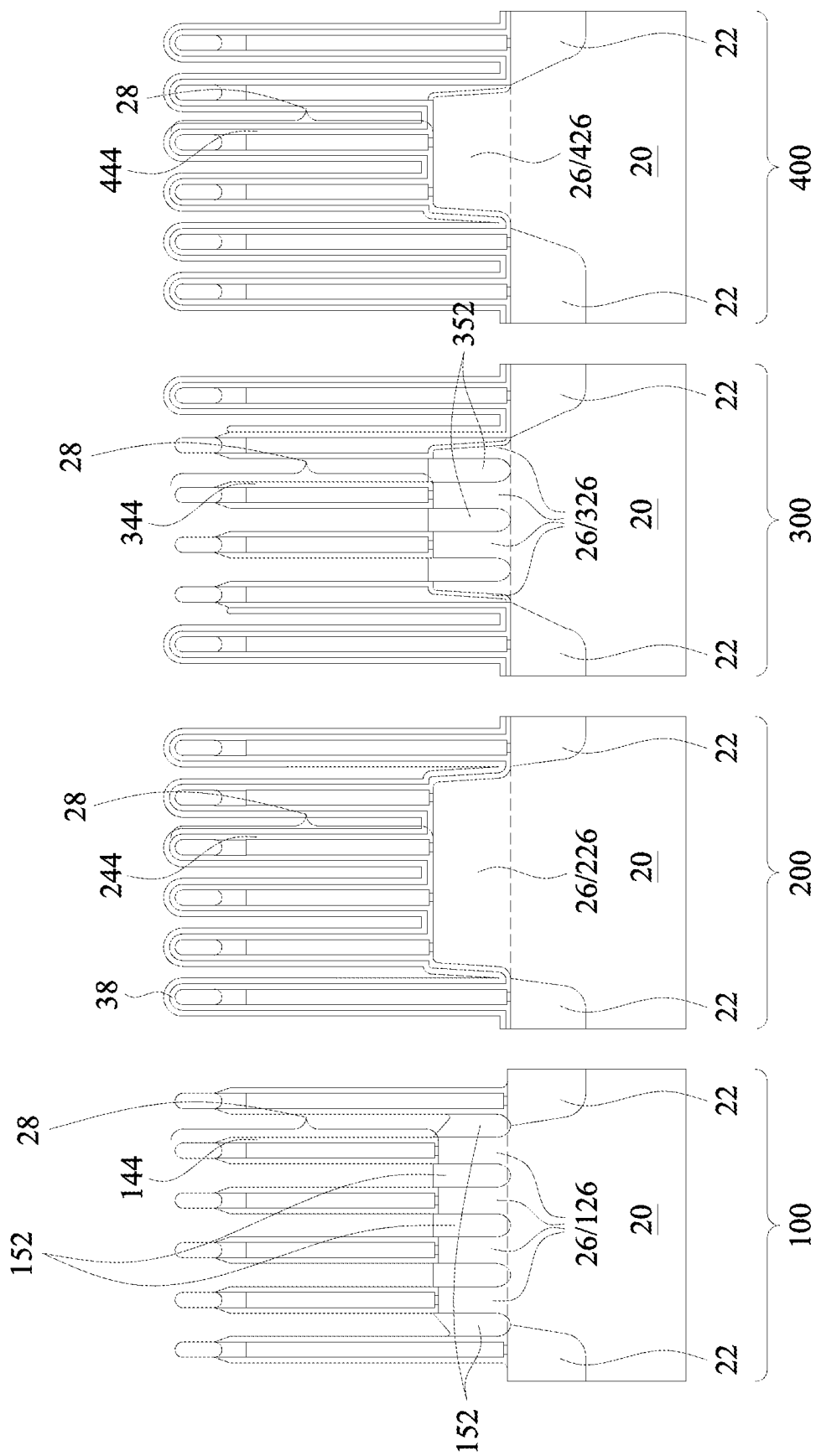


FIG. 4C

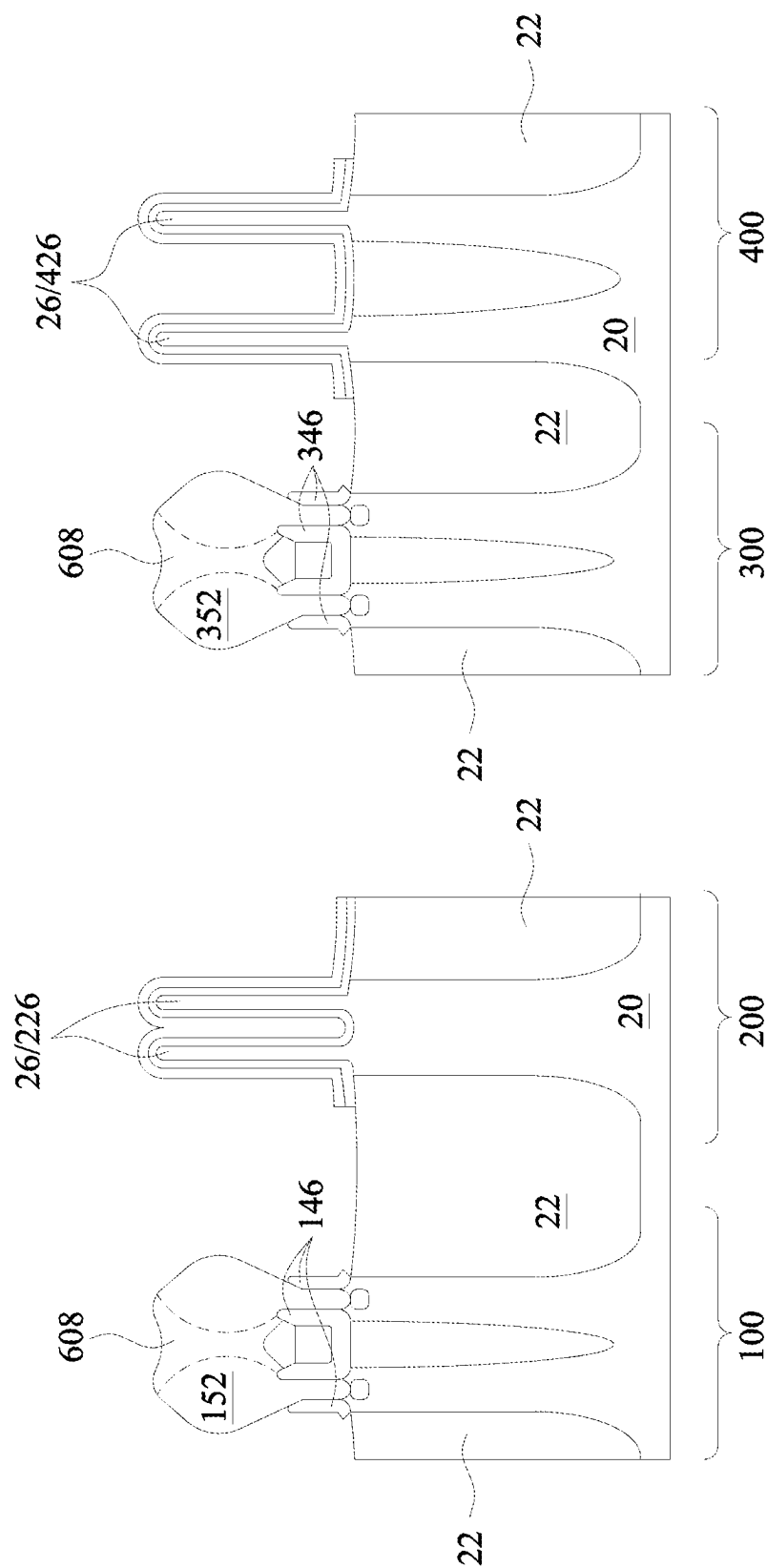


FIG. 4D

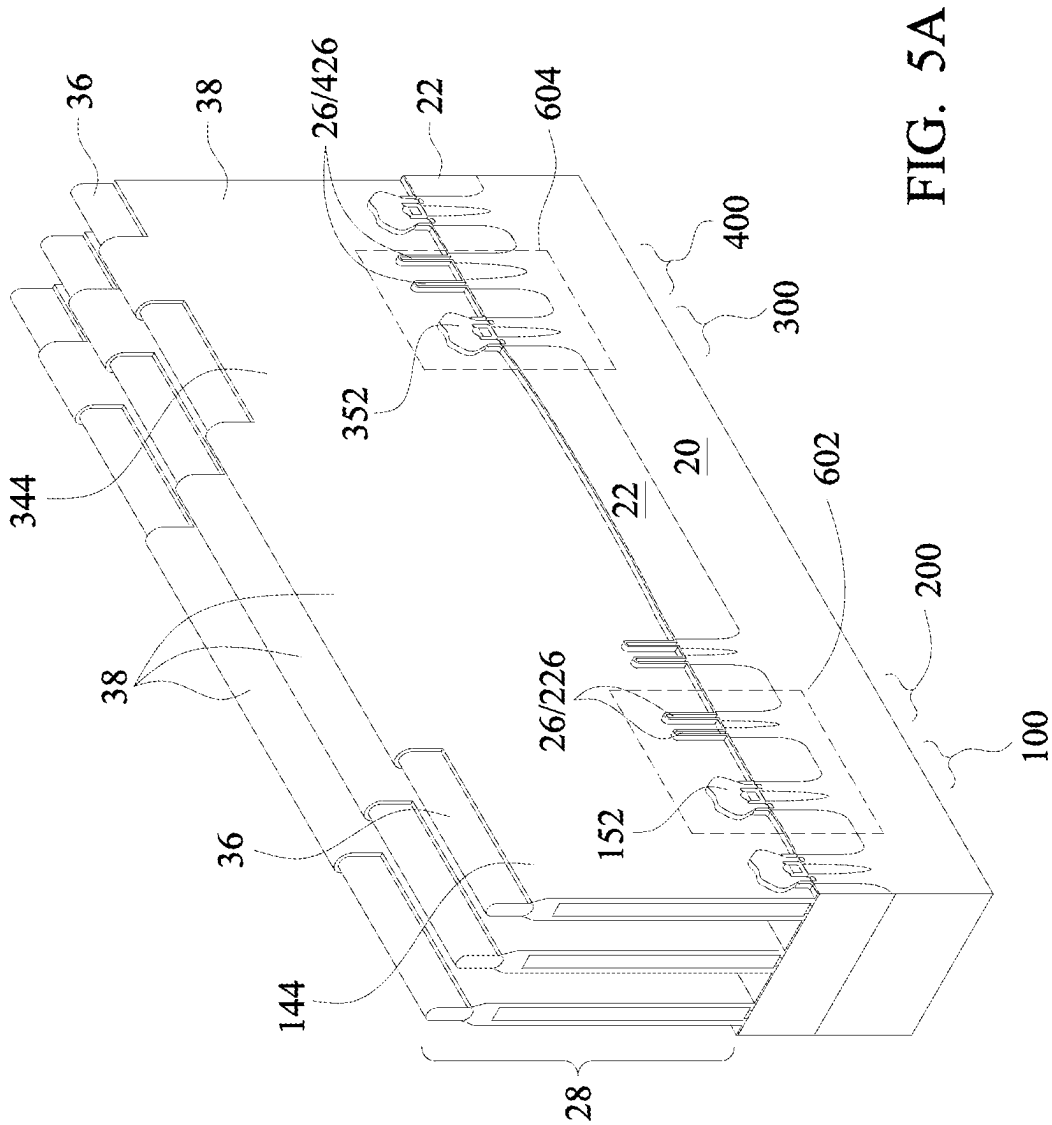


FIG. 5A

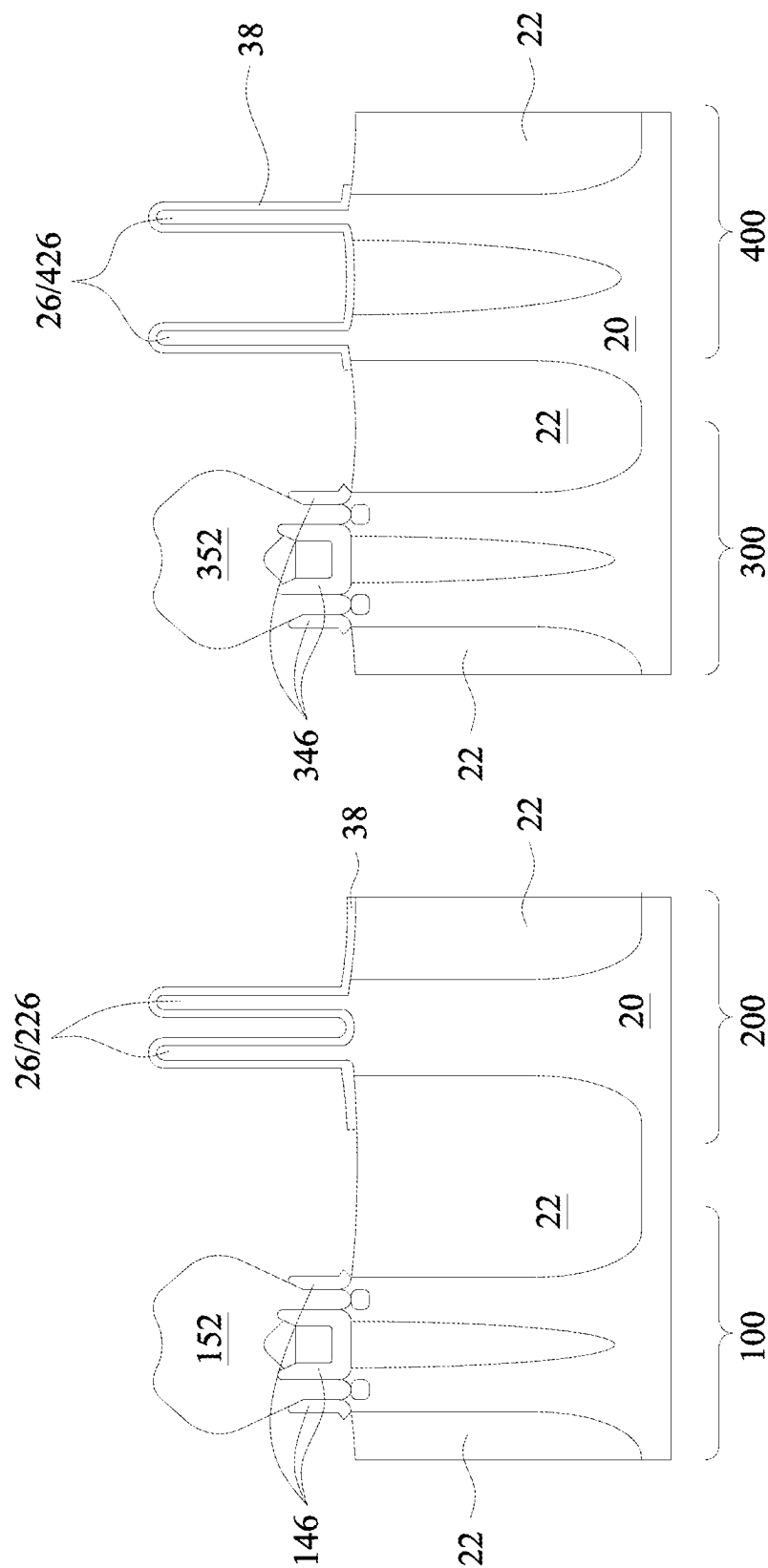


FIG. 5B

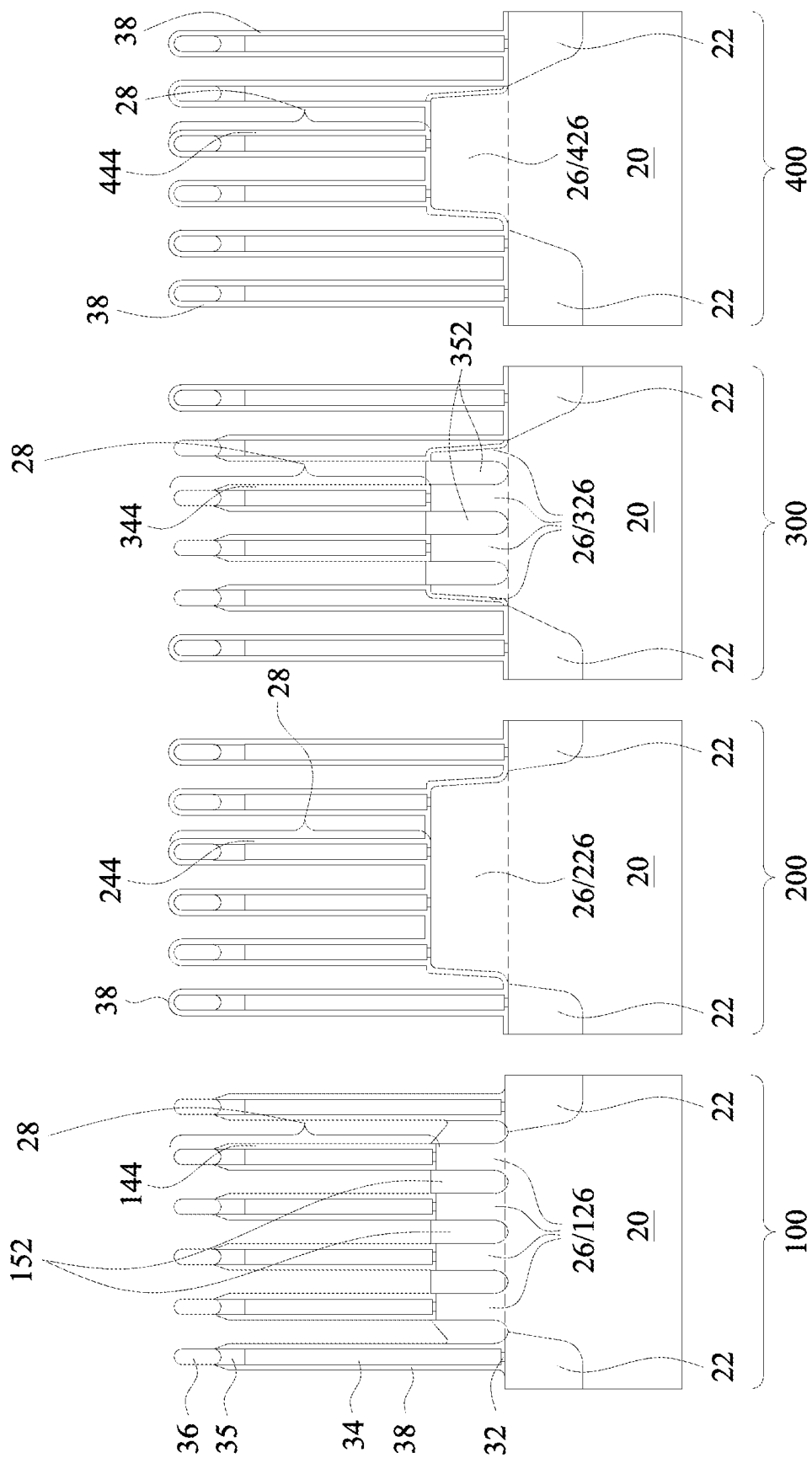


FIG. 5C

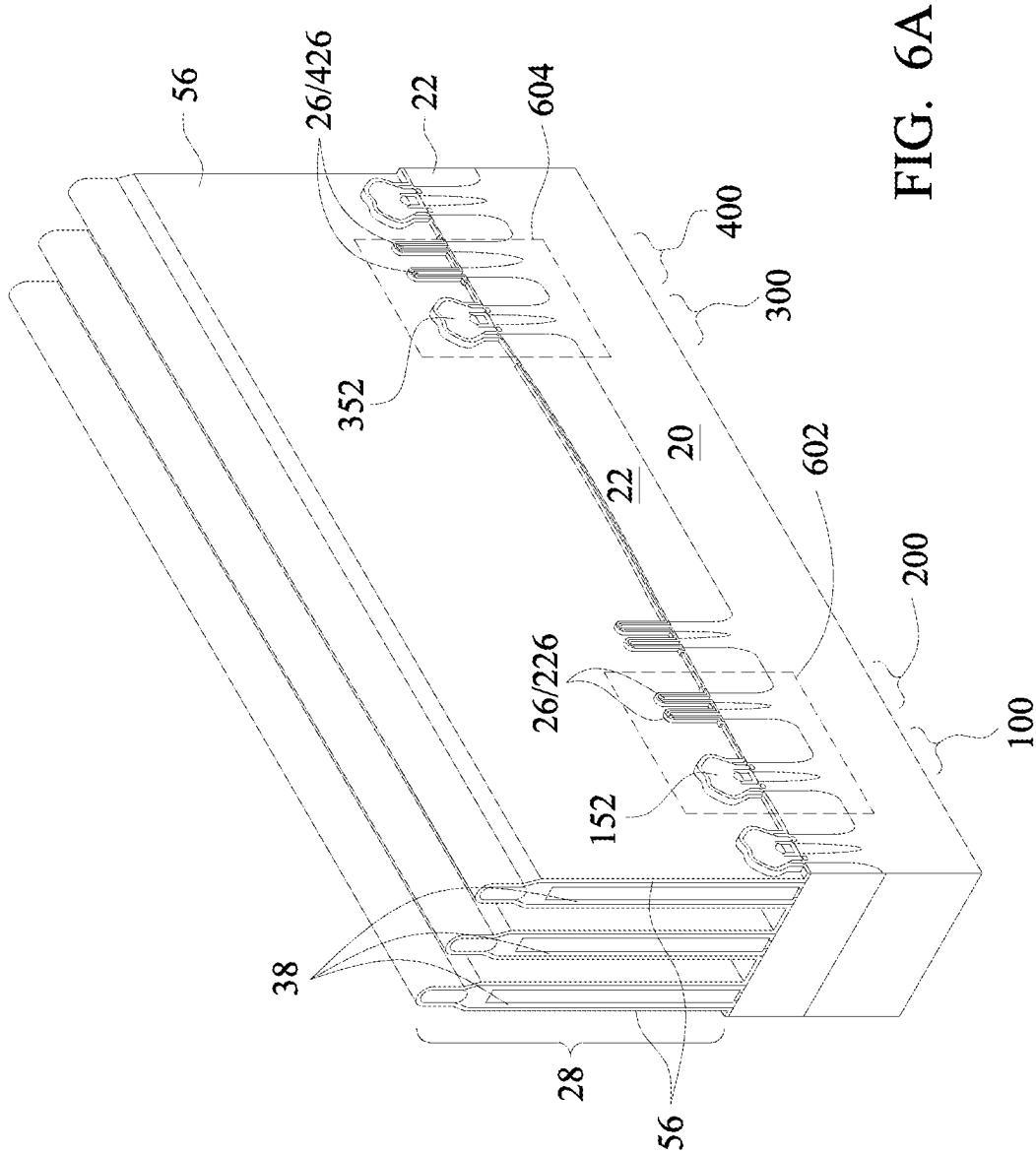


FIG. 6A

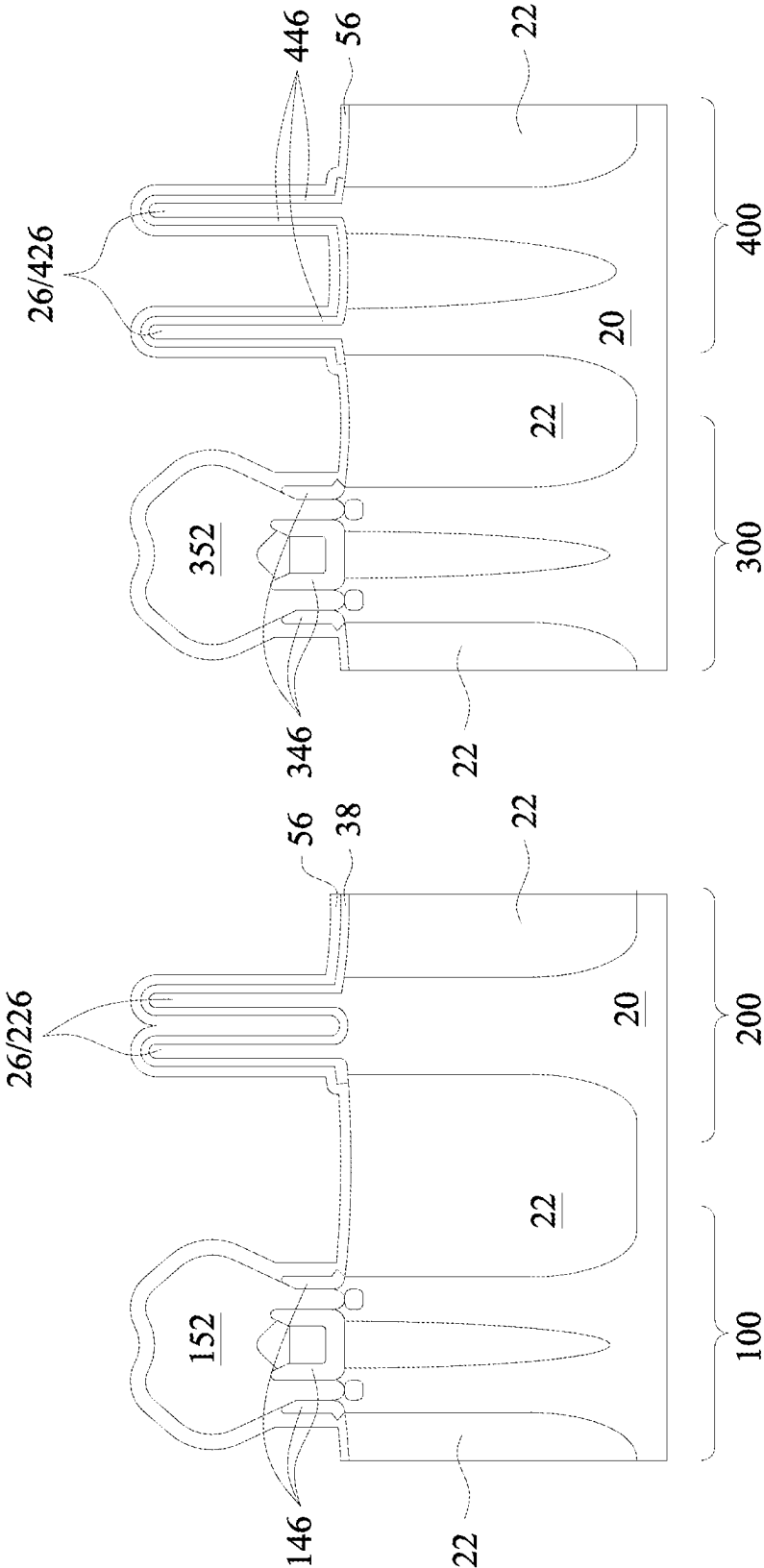


FIG. 6B

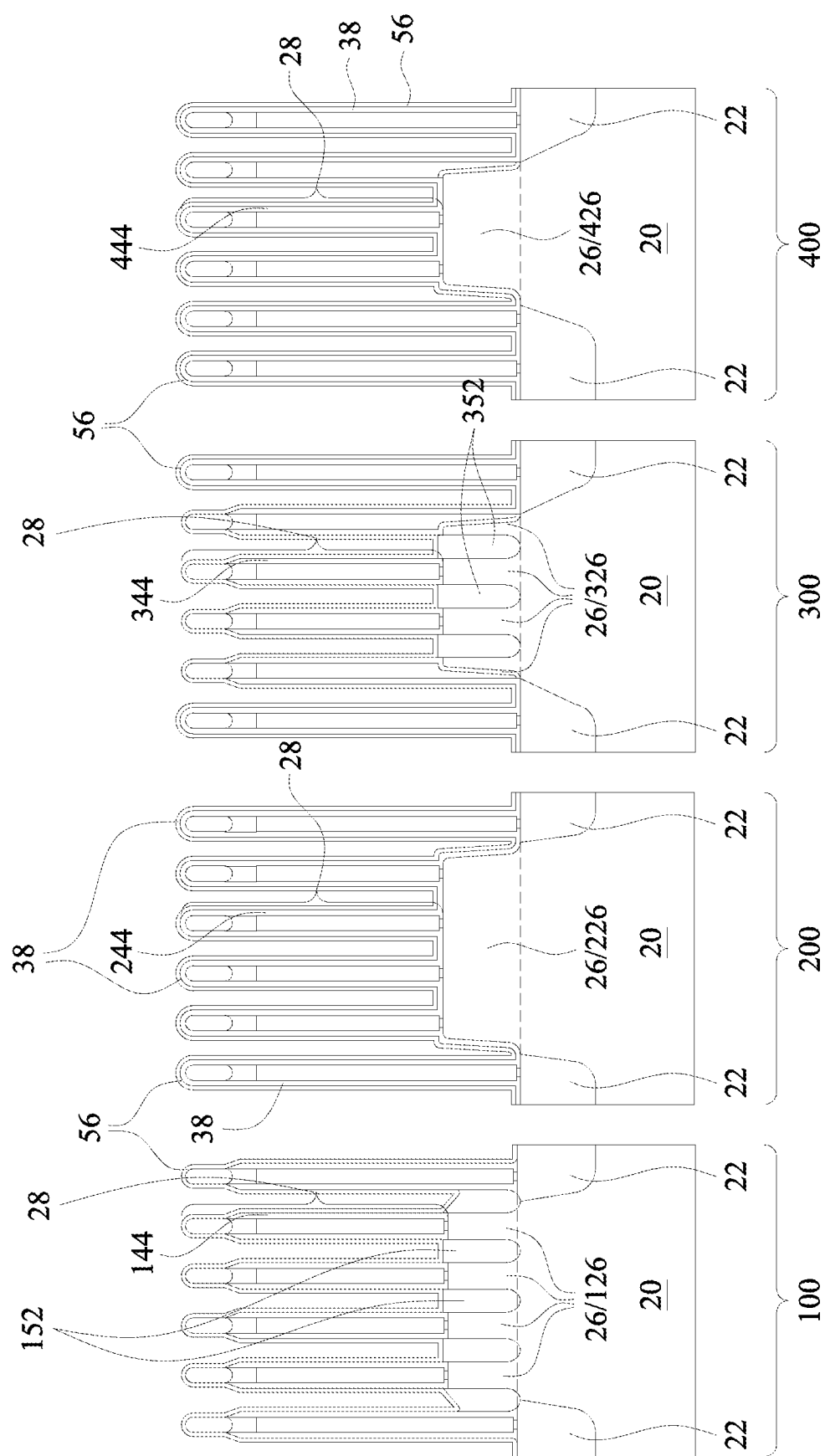


FIG. 6C

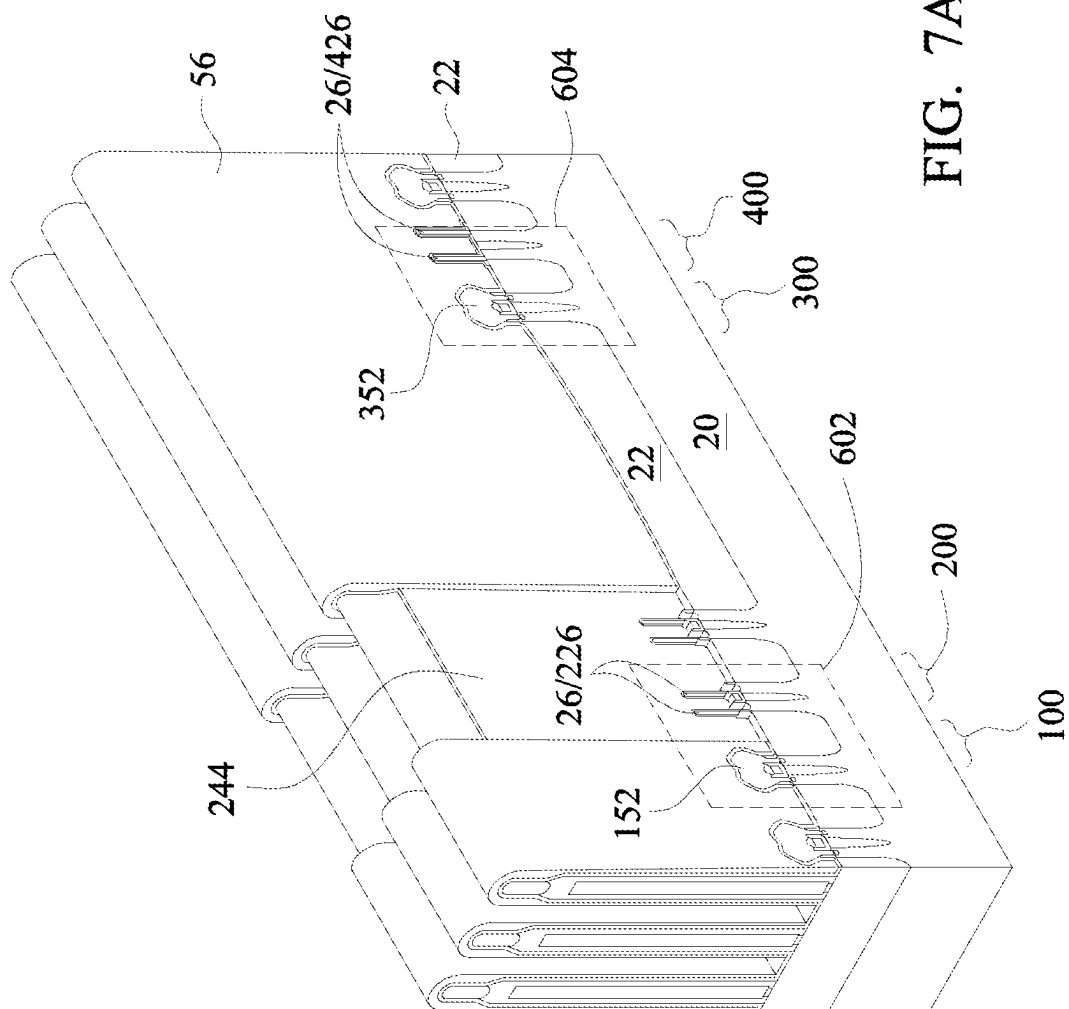


FIG. 7A

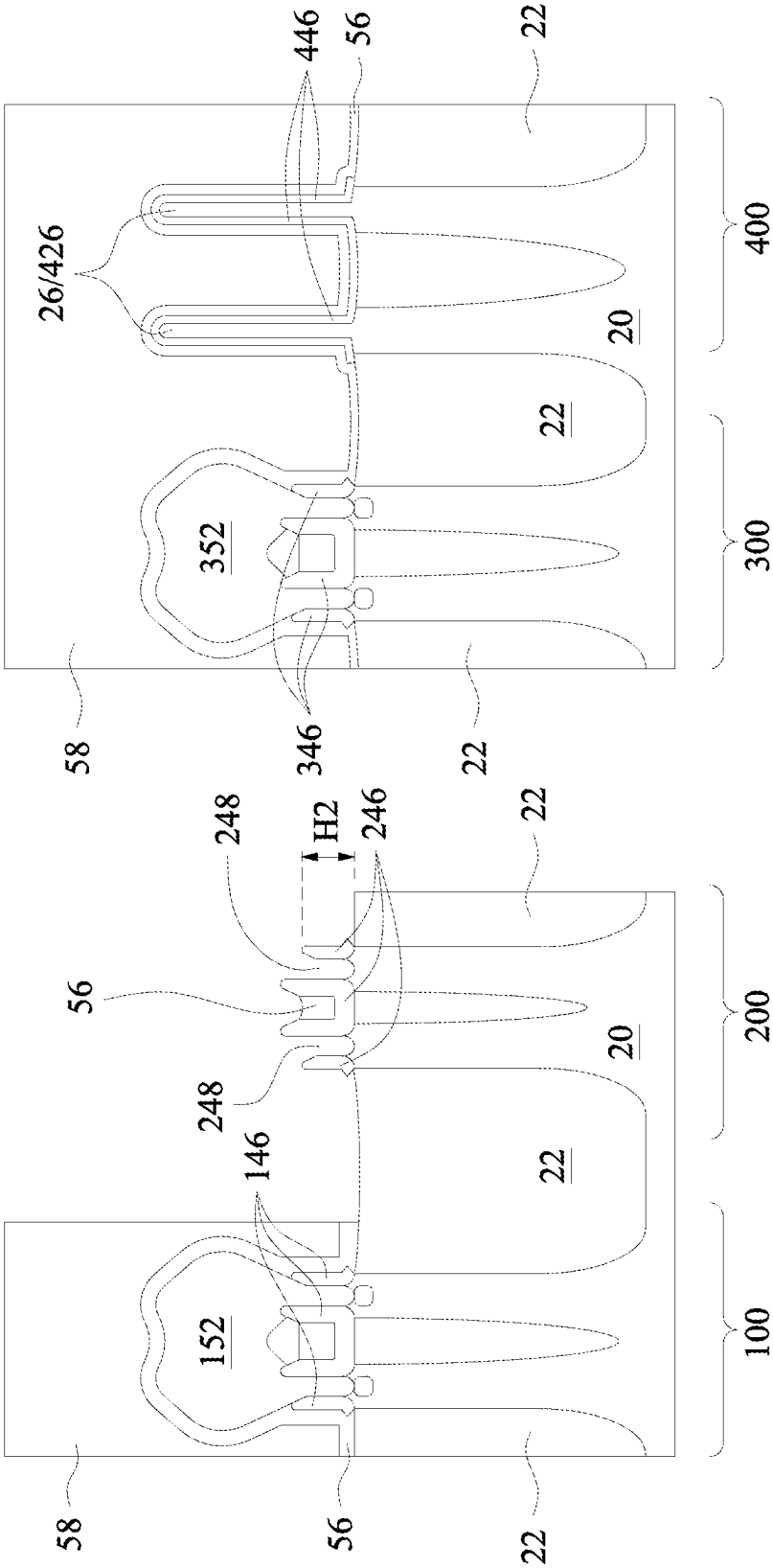


FIG. 7B

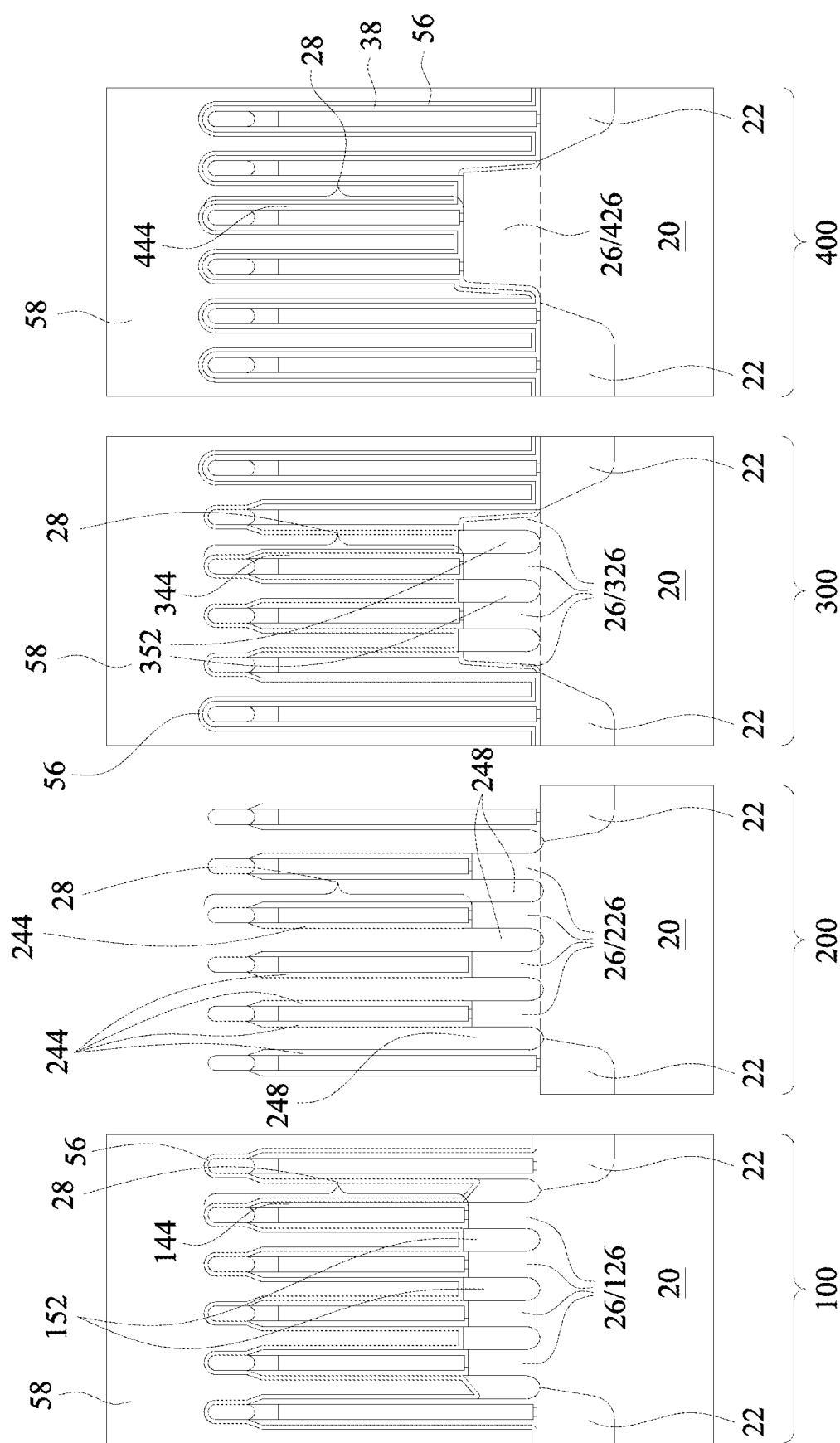


FIG. 7C

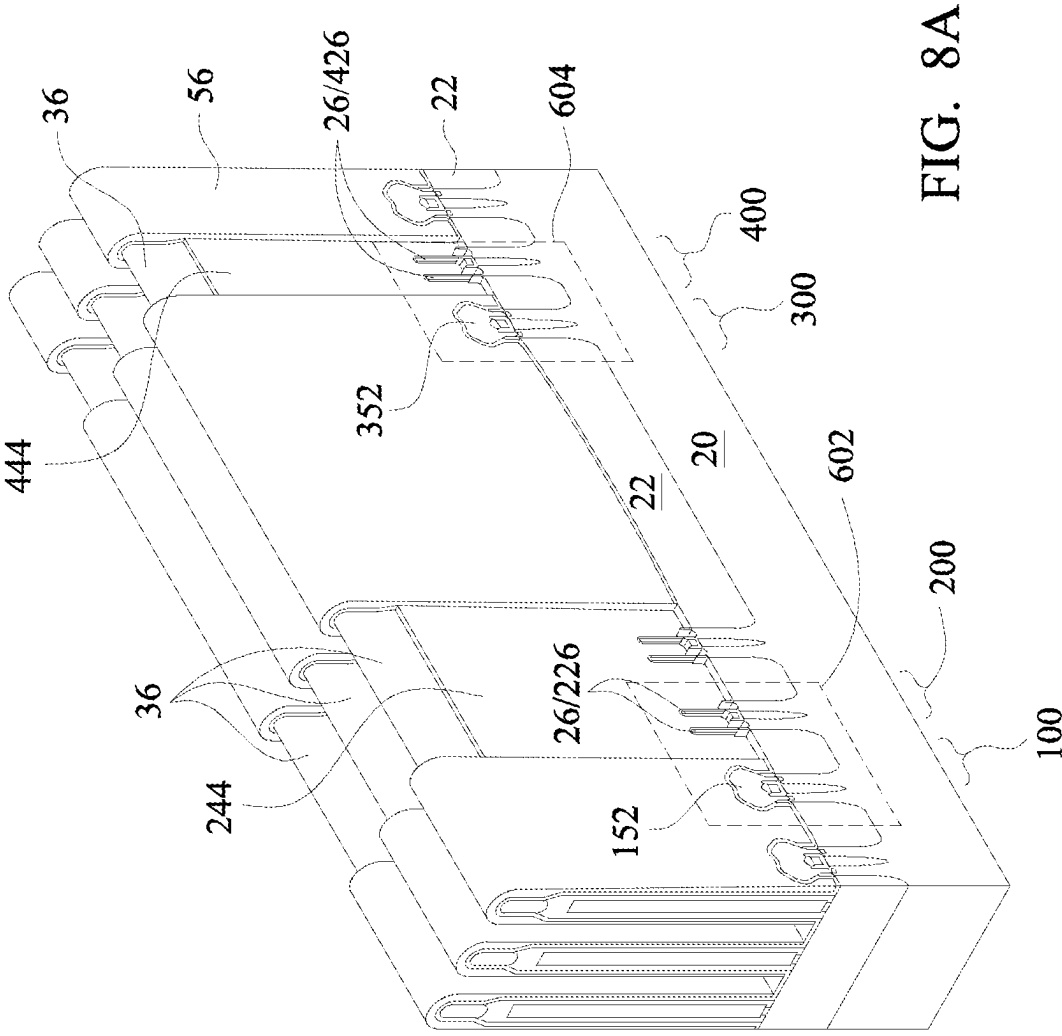


FIG. 8A

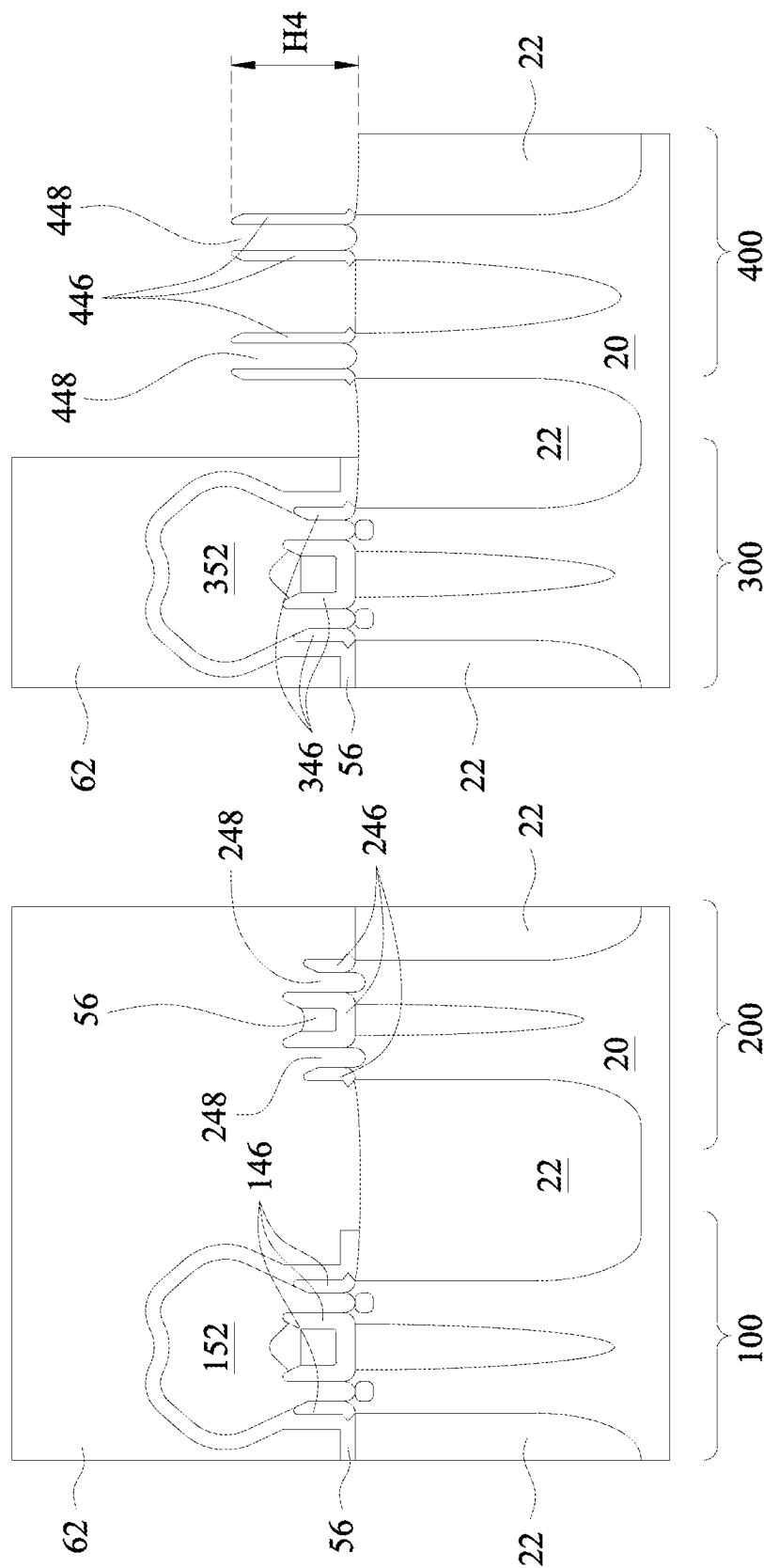


FIG. 8B

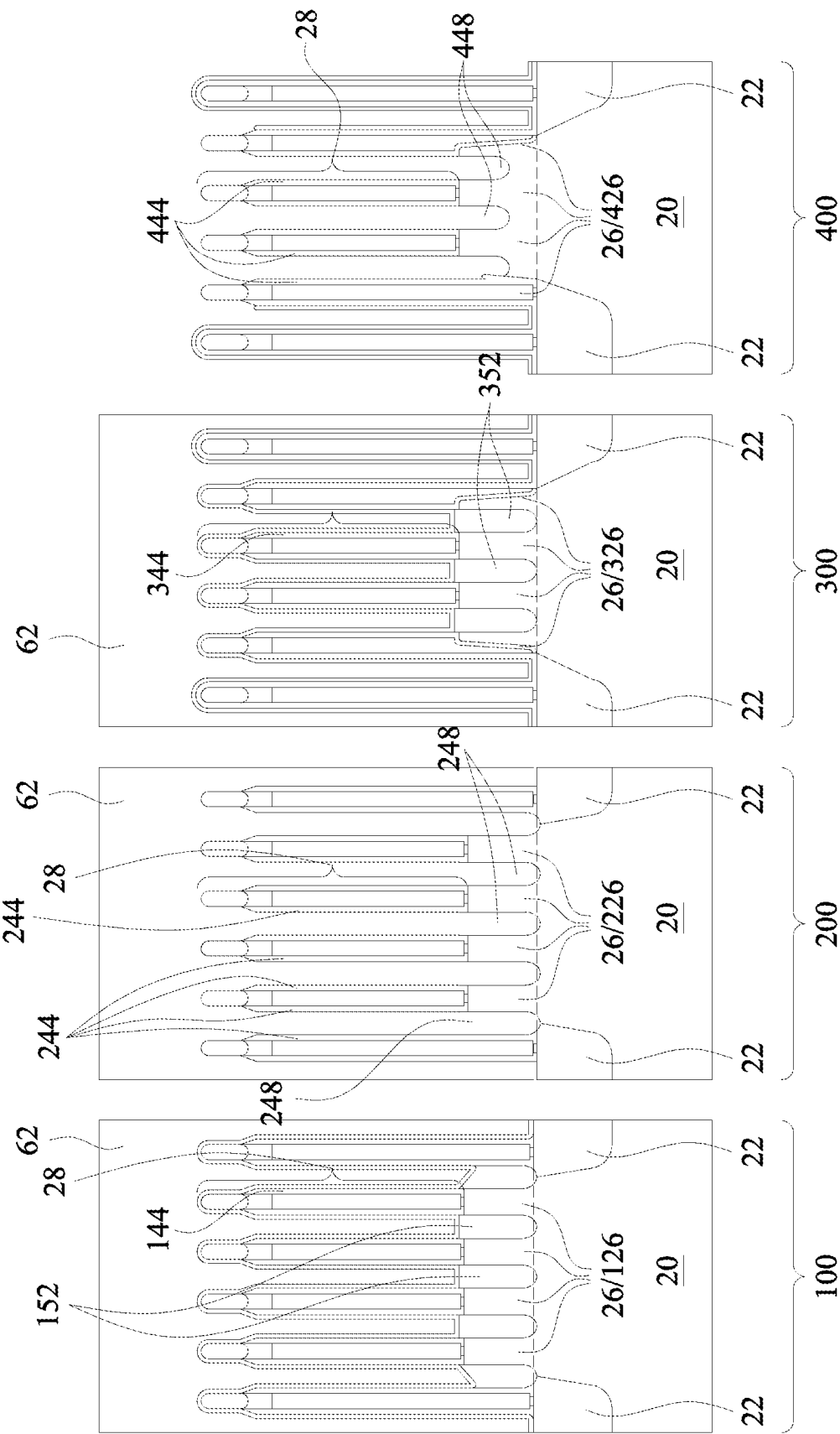


FIG. 8C

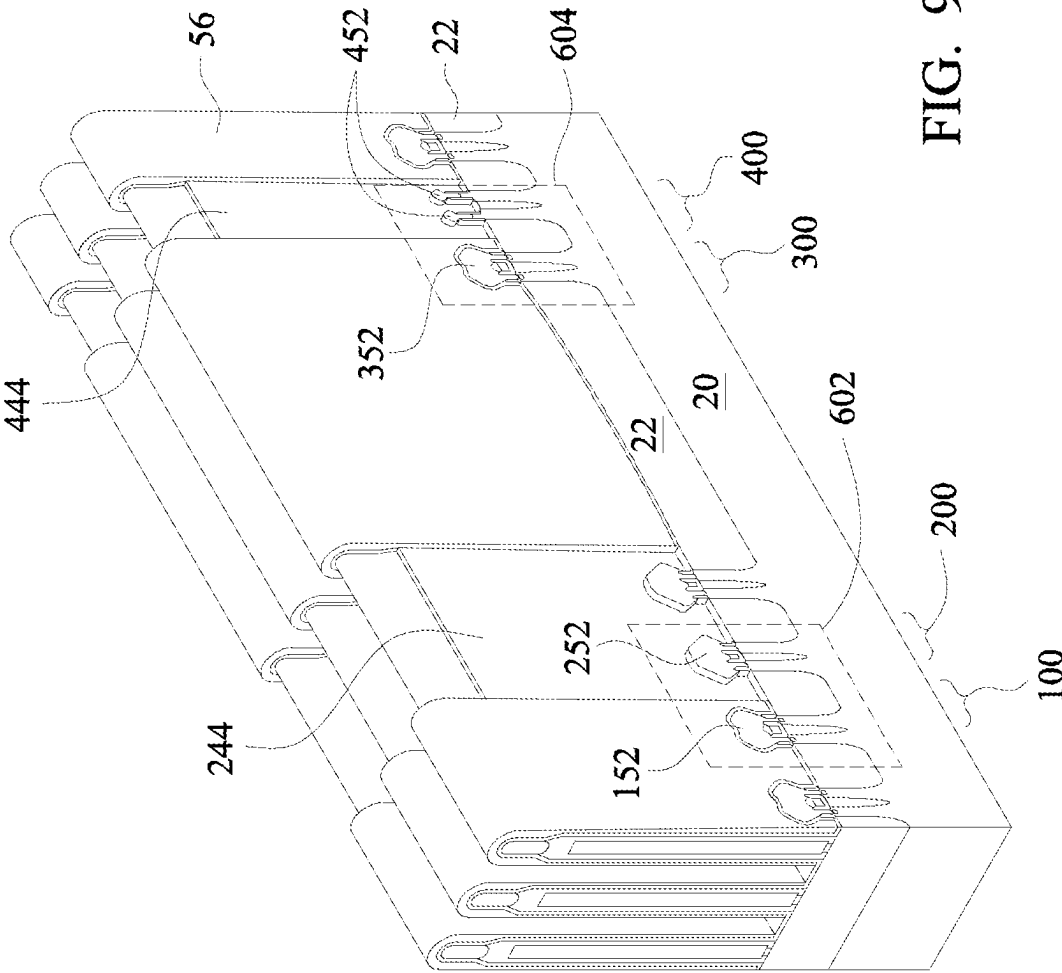


FIG. 9A

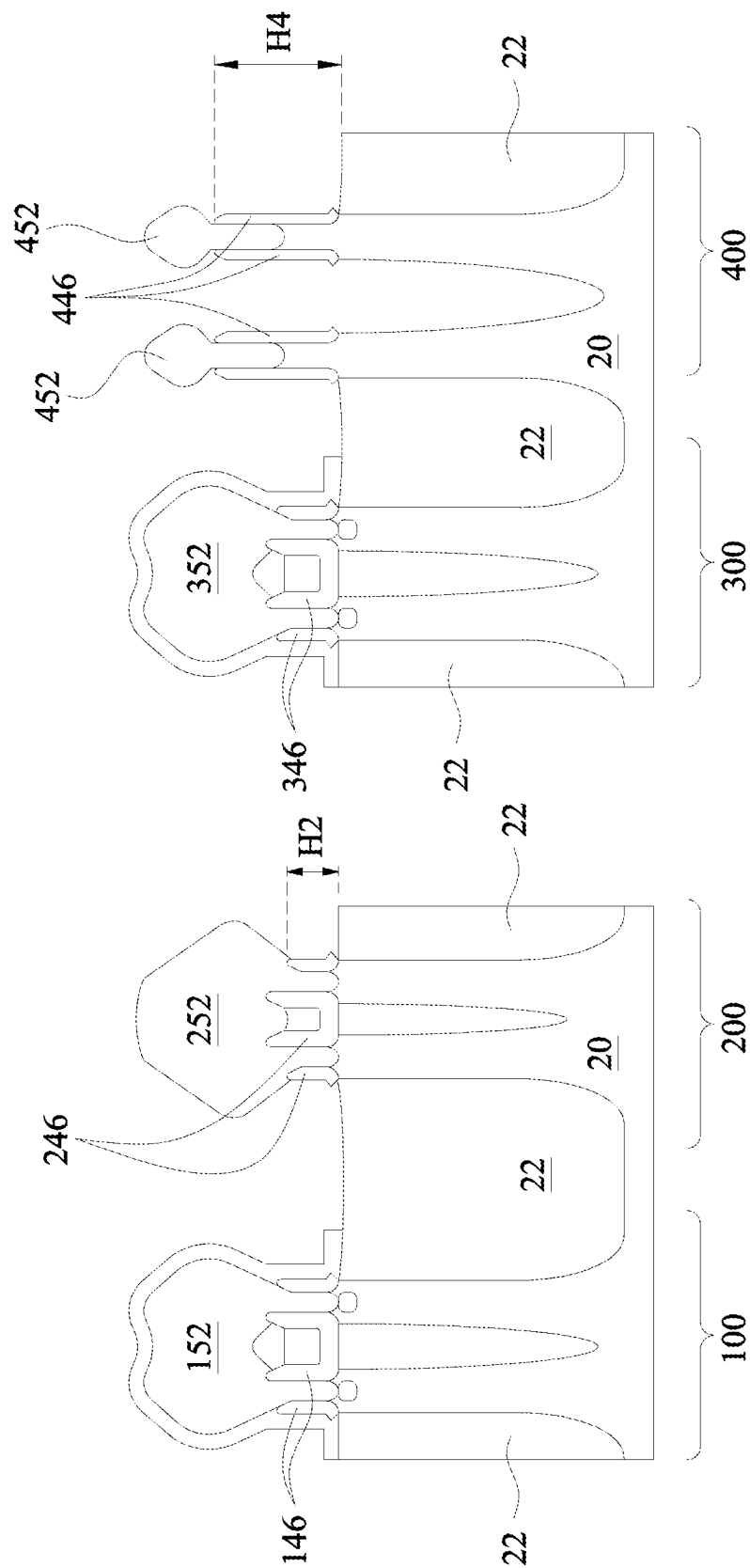


FIG. 9B

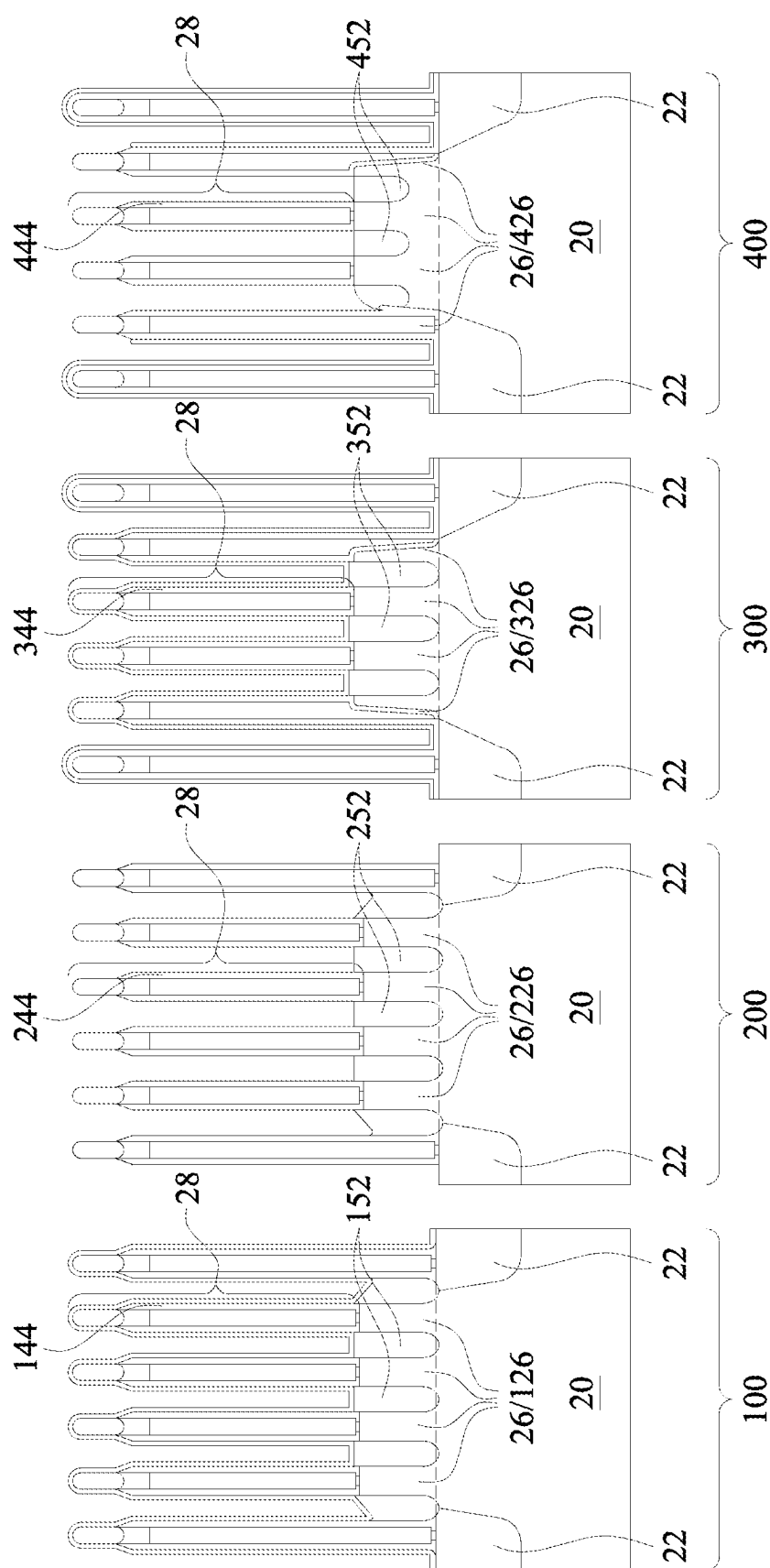


FIG. 9C

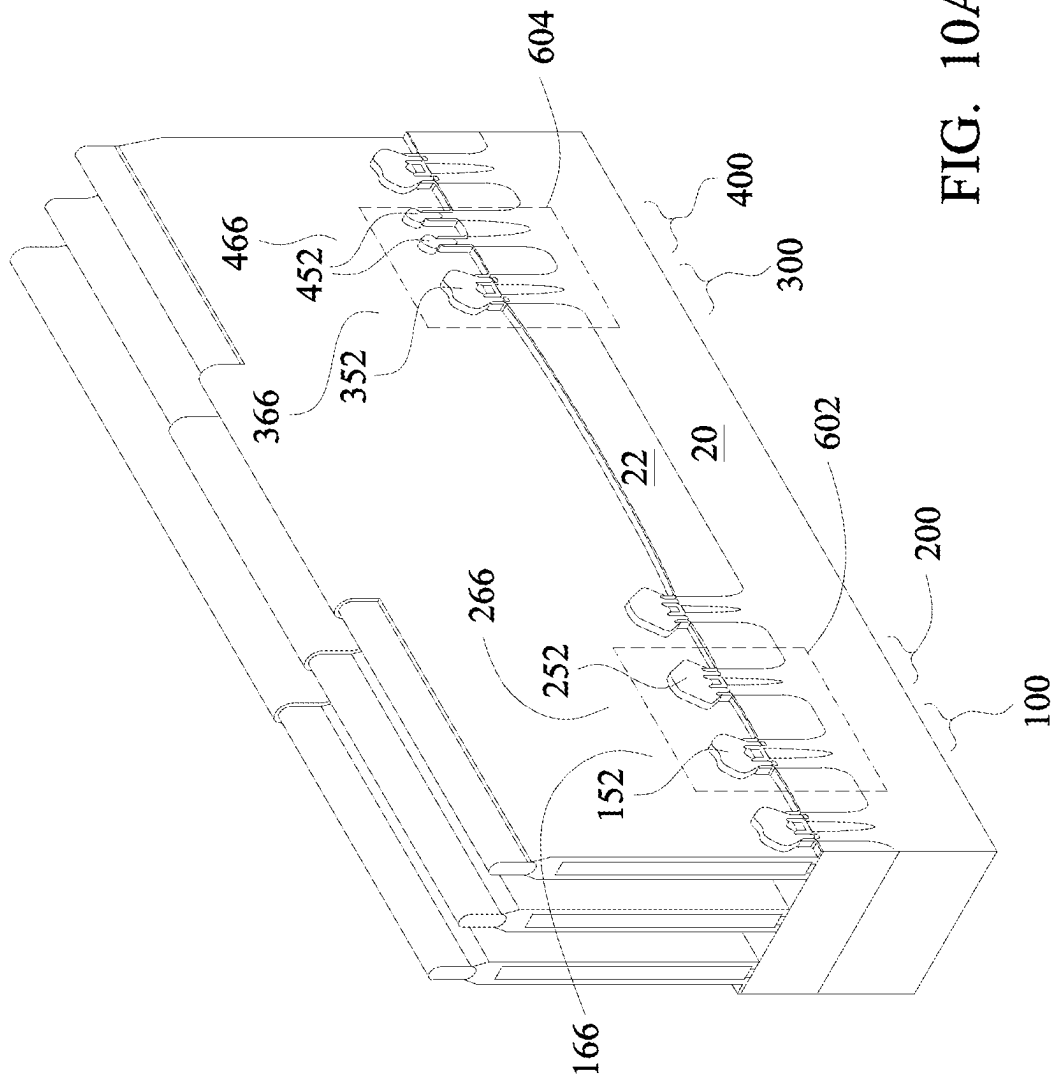


FIG. 10A

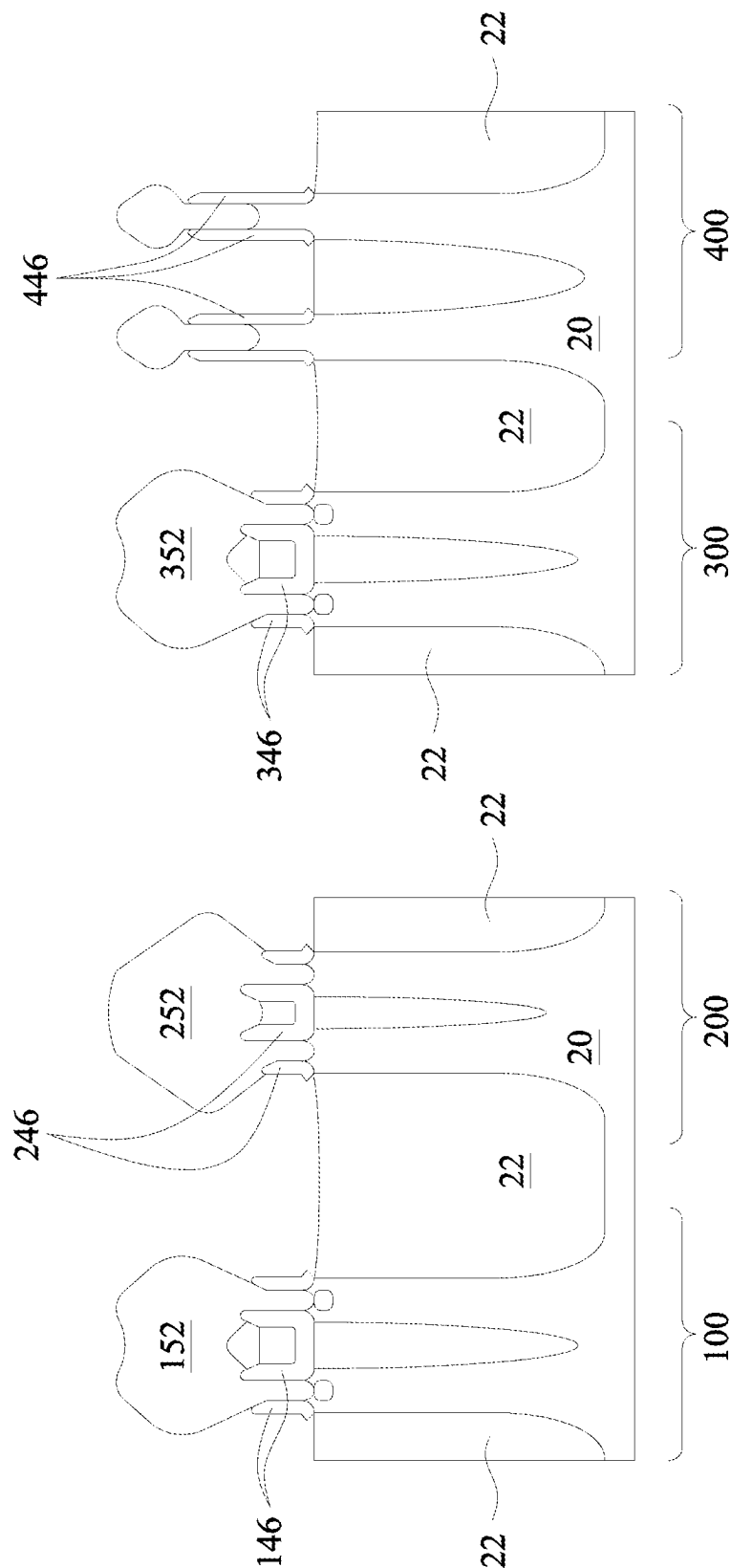


FIG. 10B

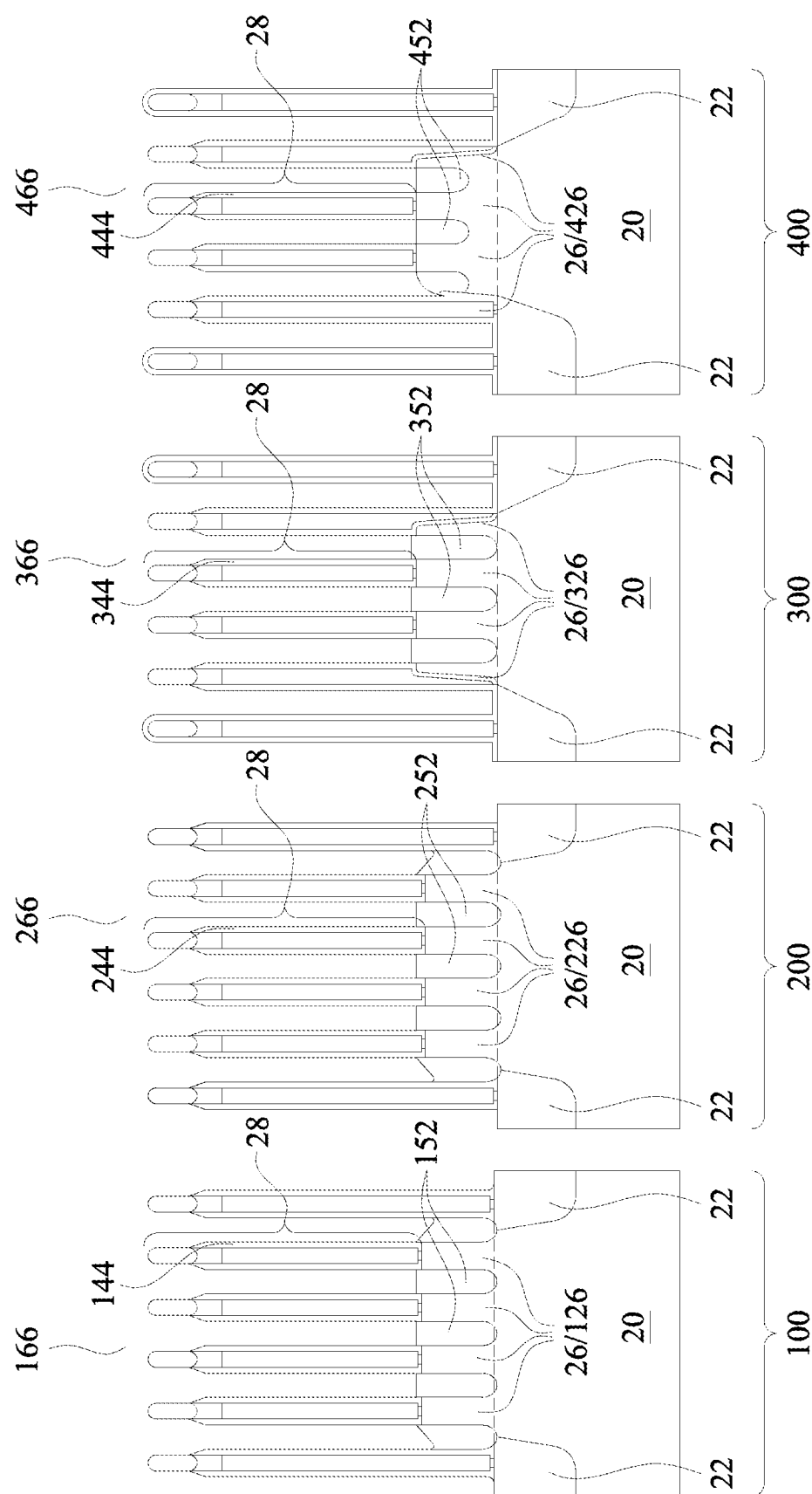


FIG. 10C

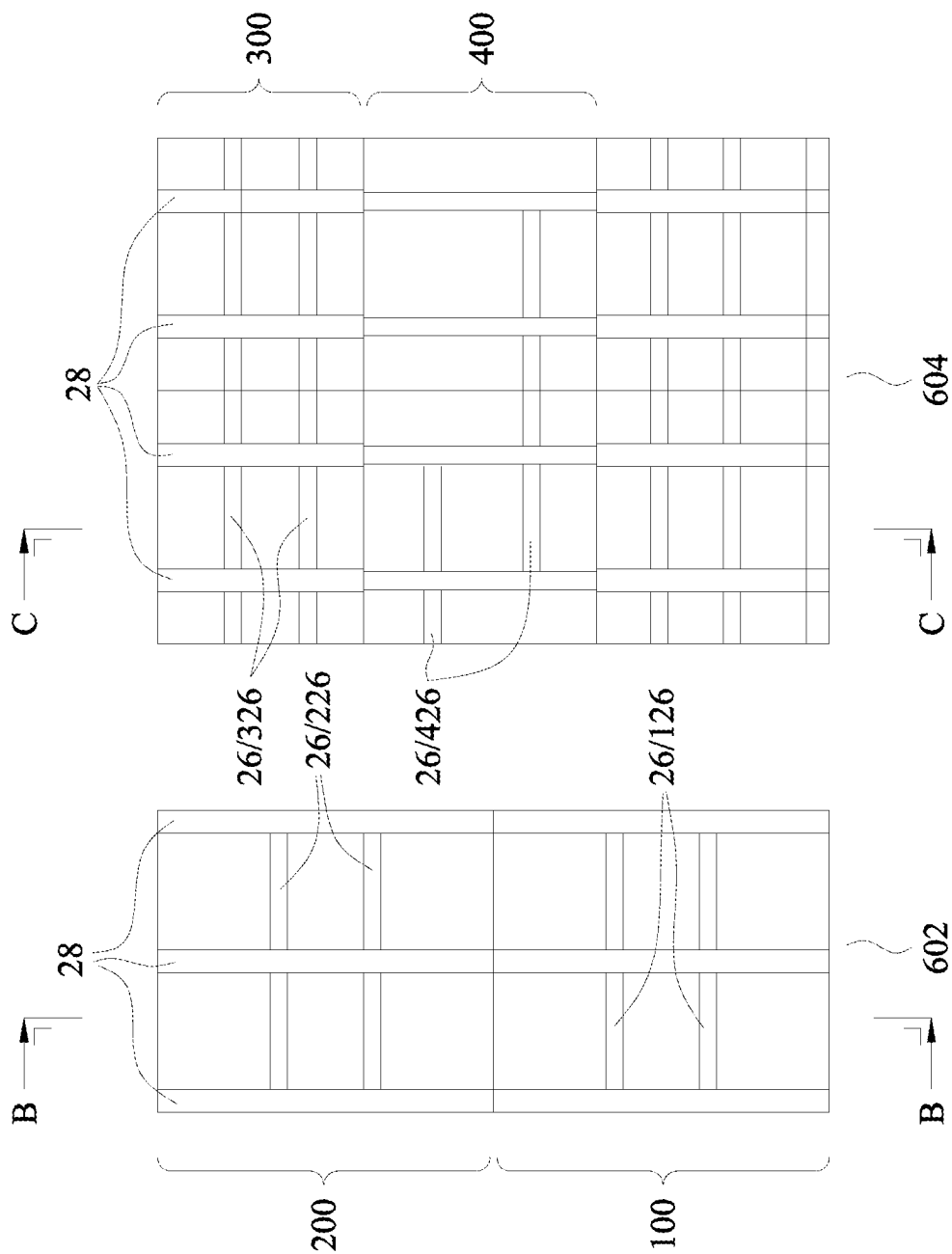
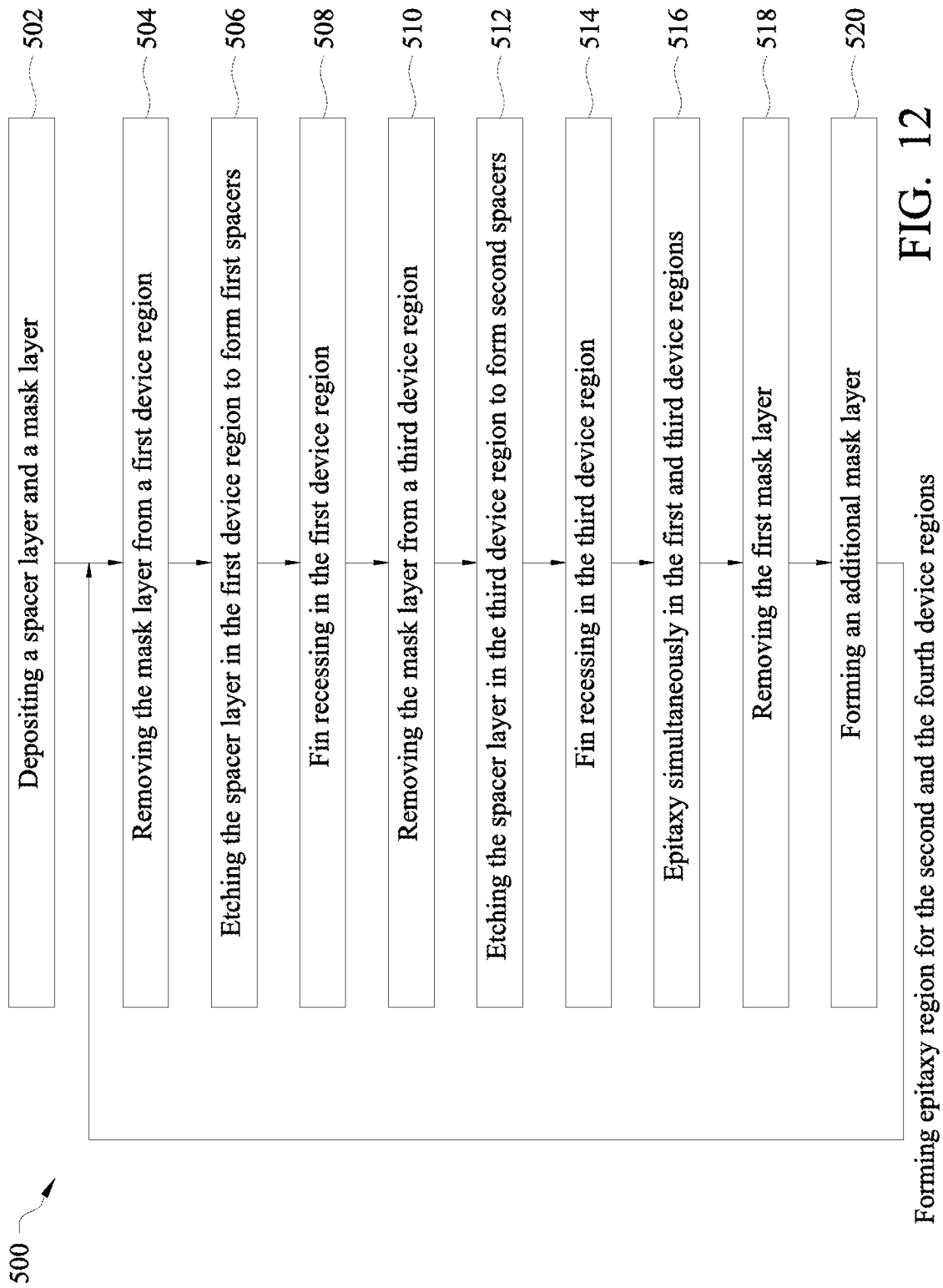


FIG. 11



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FLEXIBLE MERGE SCHEME FOR SOURCE/DRAIN EPITAXY REGIONS

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a continuation of U.S. patent application Ser. No. 17/234,201, entitled “Flexible Merge Scheme for Source/Drain Epitaxy Regions,” filed on Apr. 19, 2021, now U.S. Pat. No. 11,856,743, issued Dec. 26, 2023, which is a continuation of U.S. patent application Ser. No. 16/669,736, entitled “Flexible Merge Scheme for Source/Drain Epitaxy Regions,” filed on Oct. 31, 2019, now U.S. Pat. No. 10,985,167, issued Apr. 20, 2021, which is a continuation of U.S. patent application Ser. No. 16/206,183, entitled “Flexible Merge Scheme for Source/Drain Epitaxy Regions,” filed on Nov. 30, 2018, now U.S. Pat. No. 10,529,725 issued Jan. 7, 2020, which is a divisional of U.S. patent application Ser. No. 15/492,142, entitled “Flexible Merge Scheme for Source/Drain Epitaxy Regions,” filed on Apr. 20, 2017, now U.S. Pat. No. 10,483,266 issued Nov. 19, 2019, which applications are incorporated herein by reference.

BACKGROUND

Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs.

Such scaling down has also increased the complexity of processing and manufacturing ICs and, for these advances to be realized, similar developments in IC processing and manufacturing are needed. For example, three-dimensional transistors such as a Fin Field-Effect Transistors (FinFETs) have been introduced to replace planar transistors. Although existing FinFET devices and methods of fabricating FinFET devices have been generally adequate for their intended purposes, they have not been entirely satisfactory in all respects. For example, the FinFETs for different circuits such as core (logic) circuits and Static Random Access Memory (SRAM) circuits may have different designs, and the source/drain epitaxy regions grown from neighboring fins may need to be merged for some circuits (such as logic circuits), and need to be separated from each other for other circuits (such as SRAM circuits). However, to save manufacturing cost, the epitaxy for different regions is performed simultaneously. This causes difficulty for selectively making epitaxy regions merged for some circuits, and not merged for other circuits. Accordingly, the merged epitaxy regions need to be trimmed to separate the merged epitaxy regions from each other.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

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FIGS. 1A through 10C are cross-sectional views and perspective views of intermediate stages in the formation of Fin Field-Effect Transistors (FinFETs) in accordance with some embodiments.

FIG. 11 illustrates the exemplary layouts of a logic circuit and a Static Random Access Memory (SRAM) circuit in accordance with some embodiments.

FIG. 12 illustrates a process flow for forming FinFETs in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “underlying,” “below,” “lower,” “overlying,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Fin Field-Effect Transistor (FinFETs) and the method of forming the same are provided in accordance with various exemplary embodiments. The intermediate stages in the formation of the FinFETs are illustrated. The variations of some embodiments are discussed. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements.

FIGS. 1A through 10C illustrate the intermediate stages in the formation of FinFETs. The steps shown in FIGS. 1A through 10C are also illustrated schematically in the process flow 500 shown in FIG. 12. Each of the figure numbers in FIGS. 1A through 10C may include letter “A,” “B,” or “C,” wherein letter “A” indicates that the respective figure illustrates a perspective view, and letter “B” indicates that the respective figure is obtained from the plane same as the vertical plane containing line B-B in FIG. 1A, and letter “C” indicates that the respective figure is obtained from (and combined) the planes same as the vertical planes containing lines C-C in FIG. 1A. Accordingly, the figures whose numbers include letter “B” show the cross-sectional views obtained from the vertical planes parallel to the lengthwise directions of gate stacks, and the figures whose numbers include letter “C” show the cross-sectional views obtained from the vertical planes parallel to the lengthwise directions of semiconductor fins, which will be discussed in detail in subsequent paragraphs.

FIG. 1A illustrates a perspective view in the formation of a structure including substrate **20**, isolation regions **22**, semiconductor strips **24** between isolation regions **22**, and semiconductor fins **26** over the top surfaces of isolation regions **22**. Substrate **20** is a semiconductor substrate, which may be a silicon substrate, a silicon carbon substrate, or a substrate formed of other semiconductor materials such as III-V compound semiconductor materials. Substrate **20** may be lightly doped with a p-type or an n-type impurity.

Isolation regions **22** may be, for example, Shallow Trench Isolation (STI) regions. The formation of STI regions **22** may include etching semiconductor substrate **20** to form trenches, and filling the trenches with a dielectric material(s) to form STI regions **22**. STI regions **22** may include silicon oxide, and other dielectric materials such as nitrides may also be used. Semiconductor fins **26** overlap the underlying semiconductor strips **24**. The formation of semiconductor fins **26** may include recessing STI regions **22**, so that the portions of semiconductor material between the removed portions of STI regions **22** become semiconductor fins **26**. Semiconductor fins **26** and some or substantially entireties of semiconductor strips **24** may be formed of silicon (with no germanium therein) or other silicon-containing compound including, and not limited to, silicon carbon, silicon germanium, or the like.

A plurality of parallel gate stacks **28** is formed on semiconductor fins **26**. Gate stacks **28** are parallel to each other, and cover portions of semiconductor fins **26**, while leaving some other portions of semiconductor fins **26** uncovered. Gate stacks **28** include gate dielectrics **32** on the sidewalls and the top surfaces of semiconductor fins **26**, and gate electrodes **34** over gate dielectrics **32**. Gate dielectrics **32** may be selected from silicon oxide, silicon nitride, gallium oxide, aluminum oxide, scandium oxide, zirconium oxide, lanthanum oxide, hafnium oxide, combinations thereof, and multi-layers thereof. Gate electrodes **34** may be formed of a conductive material that includes polysilicon, a refractory metal, or the respective compound including, e.g., polysilicon, Ti, W, TiAl, TaC, TaCN, TaAlC, TaAlCN, TiN, and TiW. In other examples, gate electrodes **34** include nickel (Ni), gold (Au), copper (Cu), or the alloys thereof.

In accordance with some embodiments of the present disclosure, gate stacks **28** remain in the final FinFETs, and form the gate stacks of the final FinFETs. In accordance with alternative embodiments of the present disclosure, gate stacks **28** are dummy gate stacks that will be replaced by replacement gates in subsequent steps. Accordingly, gate stacks **28** may include dummy gate electrodes (which is also denoted as **34**), which may comprise polysilicon, for example. Dummy gate dielectrics **32** may, or may not, be formed between dummy gate electrodes **34** and semiconductor fins **26**.

Gate stacks **28** may also include hard masks **35** and **36** formed over gate electrodes **34**. In accordance with some embodiments, hard masks **35** are formed of silicon oxide, silicon oxycarbo-nitride (SiOCN), or the like. Hard masks **36** may be formed of silicon nitride (SiN), SiOCN, SiOC, or other dielectric materials in accordance with some embodiments.

The perspective view show in FIG. 1A illustrates the exemplary layouts of circuits **602** and **604** as schematically represented by dashed boxes. In accordance with some embodiments, each of circuits **602** and **604** is selected from logic circuits or SRAM circuits, and circuits **602** and **604** may be same types of circuits or different types of circuits. In the following discussion, circuits **602** and **604** are referred

to as a logic circuit and an SRAM circuit, respectively as an example, while other combinations are also contemplated.

In accordance with some embodiments, circuit **602** is formed in a device region including n-type FinFET region **200** and p-type FinFET region **200**, and circuit **604** is formed in a device region including n-type FinFET region **300** and p-type FinFET region **400**. Device regions **100**, **200**, **300**, and **400** are also illustrated in FIGS. 1B and 1C through 10C. Semiconductor fins **126**, **226**, **326**, and **426** are formed in regions **100**, **200**, **300**, and **400**, respectively, and are referred to collectively as semiconductor fins **26**. Gate stacks **28** are formed in the directions perpendicular to the lengthwise directions of semiconductor fins **26**. It is noted that although the gate stacks **28** are illustrated as continuously extending into different device regions **100**, **200**, **300**, and **400** for a compact illustration purpose, the gate stacks **28** in different device regions may be physically separated from each other, or some gate stacks **28** in some of device regions may be connected in any combination, while the gate stacks **28** in other device regions are separated.

FIG. 1B illustrates the cross-sectional views of semiconductor fins **26** in device regions **100**, **200**, **300**, and **400**, wherein the cross-sectional view is obtained from the plane crossing line B-B in FIG. 1A. Also, the plane of cross-sectional view is obtained from the middle of two neighboring gate stacks **28** (as schematically illustrated in FIG. 1C). As shown in FIG. 1B, distance D1 between neighboring fins **126** may be greater than, equal to, or smaller than, distance D1' between neighboring fins **326**. Distance D2 between neighboring fins **226** may be greater than, equal to, or smaller than, distance D2' between neighboring fins **426**. The illustrated view in FIG. 1B reflects the structures shown in the regions marked by dashed lines **602** and **604** in FIG. 1A (also refer to Figure ii).

FIG. 1C illustrates the cross-sectional views of device regions **100**, **200**, **300**, and **400**, wherein the cross-sectional views are obtained from the planes crossing lines C-C in FIG. 1A.

As shown in FIGS. 1A, 1B, and 1C, dielectric layer **38** is formed. The respective step is illustrated as step **502** in the process flow shown in FIG. 12. Dielectric layer **38** is alternatively referred to as a spacer layer. In accordance with some embodiments of the present disclosure, spacer layer **38** is formed of silicon nitride, silicon oxide, silicon carbo-nitride (SiCN), silicon oxy-carbo-nitride (SiOCN), silicon oxynitride (SiON), while other dielectric materials may be used. Spacer layer **38** may have a thickness in the range between about 2 nm and about 5 nm.

Spacer layer **38** is formed as a conformal layer, and hence covers the top surfaces and the sidewalls of semiconductor fins **26** (FIG. 1B) and gate stacks **28** (FIG. 1C). The portions of spacer layer **38** on the sidewalls of semiconductor fins **26** are used to form fin spacers, as shown in FIG. 2B, and the portions of spacer layer **38** on the sidewalls of semiconductor fins **26** are used to form gate spacers.

Mask layer **40** is formed over spacer layer **38**. The respective step is also illustrated as step **502** in the process flow shown in FIG. 12. The material of mask layer **40** is selected to have a high etching selectivity with relative to the material of spacer layer **38**. In accordance with some embodiments of the present disclosure, the material of mask layer **40** is also selected from silicon nitride, silicon oxide, silicon carbo-nitride (SiCN), silicon oxy-carbo-nitride (SiOCN), and silicon oxynitride (SiON). Mask layer **40** may have a thickness in the range between about 2 nm and about 10 nm. Mask layer **40** is also formed as a conformal layer. The formation of spacer layer **38** and mask layer **40** may be

selected from conformal deposition methods such as Atomic Layer Deposition (ALD) and Chemical Vapor Deposition (CVD). Both spacer layer **38** and mask layer **40** extend into device regions **100**, **200**, **300**, and **400**.

FIGS. **2A**, **2B**, and **2C** illustrate the patterning of spacer layer **38** in region **100**. First, photo resist **42** is applied and patterned, wherein the photo resist **42** is illustrated in FIGS. **2B** and **2C**, and not in FIGS. **2A**, although it still exists in FIG. **2A**. Photo resist **42** may be a single-layer photo resist, or a tri-layer photo resist including an inorganic layer (known as middle layer) sandwiched between two photo resists (known as under layer and upper layer).

The patterned photo resist **42** covers regions **200**, **300**, and **400**, and leaves region **100** uncovered. Next, an etching step is performed to remove the portions of mask layer **40** from region **100**. The respective step is illustrated as step **504** in the process flow shown in FIG. **12**. Depending on the process and the distance between neighboring fins **126** (FIG. **1C**), mask layer **40** may or may not have a residue portion left between neighboring fins **126**, as shown in FIG. **2B**. After mask layer **40** is removed, the portion of spacer layer **38** in region **100** is exposed, and an anisotropic etching is performed to etch spacer layer **38** in region **100**, so that the top portions of spacer layer **38** on top of fins **126** are removed, exposing fins **126**. The respective step is illustrated as step **506** in the process flow shown in FIG. **12**. The remaining portions of spacer layer **38** on the sidewalls of gate stacks **28** become gate spacers **144** (FIG. **2C**), and the remaining portions of spacer layer **38** on the sidewalls of fins **126** (FIG. **1B**) become fin spacers **146** (FIG. **2B**). The etching time of spacer layer **38** is selected so that fin spacers **146** have appropriate height **H1** (FIG. **2B**).

In a subsequent step, the exposed semiconductor fins **126** are recessed, for example, in an anisotropic or isotropic etching step, so that recesses **148** (FIGS. **2B** and **2C**) are formed to extend into semiconductor fins **126**. The respective step is illustrated as step **508** in the process flow shown in FIG. **12**. The bottoms of recesses **148** may be higher than, level with, or lower than the top surfaces of STI regions **22**. The etching is performed using an etchant that attacks fins **126**, and hardly attacks fin spacers **146**. Accordingly, in the etching step, the height of fin spacers **146** is substantially not reduced. After the formation of recesses **148**, photo resist **42** is removed, for example, in an ashing step.

FIGS. **3A**, **3B**, and **3C** illustrate the patterning of spacer layer **38** in region **300**. First, photo resist **50** is applied and patterned, wherein the photo resist **50** is illustrated in FIGS. **3B** and **3C**, and not in FIGS. **3A**, although it still exists in FIG. **3A**. Photo resist **50** may also be a single-layer photo resist or a tri-layer photo resist. The patterned photo resist **50** covers regions **100**, **200**, and **400**, and leaves region **300** uncovered. Next, an etching step is performed to remove the portions of mask layer **40** in region **300**. The respective step is illustrated as step **510** in the process flow shown in FIG. **12**. Depending on the process and the distance between neighboring fins **326** (FIG. **2C**), mask layer **40** may, or may not have a residue portion left between neighboring fins **326** (FIG. **3C**). After mask layer **40** is removed, the portion of spacer layer **38** in region **300** is exposed, and an anisotropic etching is performed to etch spacer layer **38**, so that the top portions of spacer layer **38** on top of fins **326** are removed, exposing fins **326**. The respective step is illustrated as step **512** in the process flow shown in FIG. **12**. The remaining portions of spacer layer **38** on the sidewalls of gate stacks **28** become gate spacers **344** (FIG. **3C**), and the remaining portions of spacer layer **38** on the sidewalls of fins **326** (FIG. **3B**) become fin spacers **346** (FIG. **3B**). The etching time of

spacer layer **38** is selected so that fin spacers **346** will have appropriate height **H3** (FIG. **3B**).

In a subsequent step, the exposed semiconductor fins **326** are recessed, for example, in an anisotropic or isotropic etching step, so that recesses **348** (FIGS. **3B** and **3C**) are formed to extend into semiconductor fins **326**. The respective step is illustrated as step **514** in the process flow shown in FIG. **12**. The bottoms of recesses **348** may be higher than, level with or lower than the top surfaces of STI regions **22**. The etching is performed using an etchant that attacks fins **326**, and hardly attacks fin spacers **346**. Accordingly, in the etching step, the height of fin spacers **346** is substantially not reduced. After the formation of recesses **348**, photo resist **50** is removed.

FIGS. **4A**, **4B**, and **4C** illustrate the simultaneously epitaxy for forming epitaxy semiconductor regions **152** and **352** (which are source/drain regions of FinFETs) in regions **100** and **300**, respectively. The respective step is illustrated as step **516** in the process flow shown in FIG. **12**. In accordance with some embodiments of the present disclosure, the formation of epitaxy regions **152** and **352** includes epitaxially growing silicon phosphorous (SiP) or phosphorous-doped silicon carbon (SiCP), and the resulting FinFETs formed in regions **100** and **300** are n-type FinFETs. As shown in FIG. **4B**, in the initial stage of the epitaxy, the grown epitaxy regions **152** and **352** are confined by fin spacers **146** and **346**. After the epitaxy regions **152** and **352** are grown to higher than the top ends of epitaxy regions **152** and **352**, respectively, lateral growth also occurs along with the vertical growth, and epitaxy regions **152** and **352** expand laterally.

The portions of epitaxy regions **152** grown from neighboring recesses **148** may be merged as a large epitaxy region, or remain separated from each other when the epitaxy is finished. The portions of epitaxy regions **352** grown from neighboring recesses **348** may also be merged into a large epitaxy region, or remain separated from each other when the epitaxy is finished. In addition, voids **153** and **353** may be formed when mergence occurs. Whether the mergence occurs or not depends on the heights of the respective fin spacers **146** and **346**, and how long the epitaxy lasts. Accordingly, by adjusting heights **H1** and **H3** (FIG. **4B**), one of the following four scenarios may occur: the mergence occurs for both epitaxy regions **152** and **352**, the mergence occurs for epitaxy regions **152** but not for epitaxy regions **352**, the mergence occurs for epitaxy regions **352** but not for epitaxy regions **152**, and the mergence doesn't occur for either of epitaxy regions **152** and **352**. FIG. **4D** illustrates some exemplary embodiments, wherein the unmerged epitaxy regions **152** and **352** are illustrated if the corresponding epitaxy portions **608** in dashed lines don't exist.

Referring back to FIG. **4B**, if, for example, it is desirable that the mergence occur for epitaxy regions **152** but not for epitaxy regions **352**, fin spacers **146** may be formed to have height **H1** smaller than height **H3** of fin spacers **346**. As a result, the lateral expansion occurs earlier for epitaxy regions **152** than epitaxy regions **352**, and epitaxy regions **152** merge while epitaxy regions **352** don't merge. In accordance with some embodiments of the present disclosure, to make height **H1** smaller than height **H3**, the period of time **TP1** for etching spacer layer **38** (the step shown in FIG. **2B**) may be selected to be longer than the period of time **TP3** for etching spacer layer **38** (the step shown in FIG. **3B**). In accordance with some embodiments of the present disclosure, ratio **TP1/TP3** may be greater than about 1.5, and may be in the range between about 1.5 and about 5.0. As a result, height **H3/H1** may be greater than about 1.5, and may

be in the range between about 1.5 and about 5.0. With the merging of neighboring epitaxy regions, the resulting FinFET may have a higher drive (saturation current). With the neighboring epitaxy regions not merged, the resulting FinFETs may be more compact. Accordingly, different requirements of different circuits may be met at the same time without requiring the epitaxy regions to be formed by different epitaxy processes.

Converse to the above discussion, if it is desirable that the merge occur for epitaxy regions 352 but not form epitaxy regions 152, fin spacers 146 may be formed to have height H1 greater than height H3 of fin spacers 346. In accordance with some embodiments of the present disclosure, ratio TP3/TP1 may be greater than about 1.5, and may be in the range between about 1.5 and about 5.0. Also, height H1/H3 may be greater than about 1.5, and may be in the range between about 1.5 and about 5.0.

In accordance with some embodiments, after the epitaxy, an implantation is performed to implant an n-type impurity such as phosphorous or arsenic into epitaxy regions 152 and 352 to form source/drain regions, which are also referred to using reference numerals 152 and 352. In accordance with alternative embodiments, no implantation of any n-type impurity is performed, and the n-type impurity was provided through the in-situ doping occurred during the epitaxy.

FIGS. 5A through 10C illustrate the formation of epitaxy regions for the FinFETs in regions 200 and 400, wherein the respective steps are similar to the repetition of the steps for forming epitaxy regions in device regions 100 and 300, except the conductivity types of some regions are inverted. First, an etching step is performed to remove remaining portions of mask layer 40 from regions 100, 200, 300, and 400. The respective step is illustrated as step 518 in the process flow shown in FIG. 12. Some residue portions of mask layer 40 may (or may not) be left after the etch. The resulting structure is shown in FIGS. 5A, 5B, and 5C.

Next, as shown in FIGS. 6A, 6B, and 6C, mask layer 56 is formed. The respective step is illustrated as step 520 in the process flow shown in FIG. 12. The material and the formation methods of mask layer 56 may be selected from the same candidate materials and candidate methods for forming mask layer 40. For example, the material of mask layer 56 may be selected from silicon nitride, silicon oxide, SiCN, SiOCN, and SiON. Mask layer 56 may also be formed using ALD or CVD. The thickness of mask layer 56 may be in the range between about 2 nm and about 10 nm.

FIGS. 7A, 7B, and 7C illustrate the patterning of spacer layer 38 in region 200. First, photo resist 58 is applied and patterned, wherein the photo resist 58 is illustrated in FIGS. 7B and 7C, and not in FIGS. 7A, although it still exists in FIG. 7A. The patterned photo resist 58 covers regions 100, 300, and 400, and leaves region 200 uncovered. Next, an etching step is performed to remove the portions of mask layer 56 in region 200. Depending on the process and the distance between neighboring fins 226 (FIG. 2C), mask layer 56 may, or may not have a residue portion left between neighboring fins 226 (FIG. 7B). After mask layer 56 is removed, the portion of spacer layer 38 in region 200 is exposed, and an anisotropic etching is performed to etch spacer layer 38, so that the top portions of spacer layer 38 on top of fins 226 (FIG. 6B) are removed, hence exposing fins 226. The remaining portions of spacer layer 38 on the sidewalls of gate stacks 28 become gate spacers 244 (FIG. 7C), and the remaining portions of spacer layer 38 on the sidewalls of fins 226 (FIG. 7B) become fin spacers 246. The etching time of spacer layer 38 is selected so that fin spacers 246 have appropriate height H2 (FIG. 7B).

In a subsequent step, the exposed semiconductor fins 226 (FIG. 6B) are etched, for example, in an anisotropic or isotropic etching step, so that recesses 248 (FIGS. 7B and 7C) are formed to extend into semiconductor fins 226. The bottoms of recesses 248 may be higher than, level with, or lower than the top surfaces of STI regions 22. The etching is performed using an etchant that attacks fins 226, and hardly attacks fin spacers 246. Accordingly, in the etching step, the height H2 of fin spacers 246 is substantially not reduced. After the formation of recesses 248, photo resist 58 is removed.

FIGS. 8A, 8B, and 8C illustrate the patterning of spacer layer 38 in region 400. First, photo resist 62 is applied and patterned, wherein the photo resist 62 is illustrated in FIGS. 8B and 8C, and not in FIGS. 8A, although it still exists in FIG. 8A. The patterned photo resist 62 covers regions 100, 200, and 300, and leaves region 400 uncovered. Next, an etching step is performed to remove the portions of mask layer 56 in region 400. Depending on the process and the distance D2' between neighboring fins 426 (FIG. 2C), mask layer 56 may, or may not have a residue portion left between neighboring fins 426 (FIG. 8B). After mask layer 56 is removed, the portion of spacer layer 38 in region 400 is exposed, and an anisotropic etching is performed to etch spacer layer 38, so that the top portions of spacer layer 38 on top of fins 426 (FIG. 7B) are removed, exposing fins 426. The remaining portions of spacer layer 38 on the sidewalls of gate stacks 28 become gate spacers 444 (FIG. 8C), and the remaining portions of spacer layer 38 on the sidewalls of fins 426 (FIG. 8B) become fin spacers 446. The etching time of spacer layer 38 is selected so that fin spacers 446 have appropriate height H4 (FIG. 8B).

In a subsequent step, the exposed semiconductor fins 426 (FIG. 7B) are etched, for example, in an anisotropic or isotropic etching step, so that recesses 448 (FIGS. 8B and 8C) are formed to extend into semiconductor fins 426. The bottoms of recesses 448 may be higher than, level with or lower than the top surfaces of STI regions 22. The etching is performed using an etchant that attacks fins 426, and hardly attack fin spacers 446. Accordingly, in the etching step, the height of fin spacers 446 is substantially not reduced. After the formation of recesses 448, photo resist 62 is removed.

FIGS. 9A, 9B, and 9C illustrate the simultaneously epitaxy for forming epitaxy regions 252 and 452 (which are source/drain regions of FinFETs) in regions 200 and 400, respectively. In accordance with some embodiments of the present disclosure, the formation of epitaxy regions 252 and 452 includes epitaxially growing silicon germanium, wherein boron may be in-situ doped, so that the resulting FinFETs formed in regions 200 and 400 are p-type FinFETs. As also shown in FIG. 9B, in the initial stage of the epitaxy, the grown epitaxy regions 252 and 452 are confined by fin spacers 246 and 446, respectively. After epitaxy regions 252 and 452 are grown to higher than the top ends of epitaxy regions 252 and 452, respectively, lateral growth also occurs along with the vertical growth, and epitaxy regions 252 and 452 expand laterally.

The portions of epitaxy regions 252 grown from neighboring recesses 248 may be merged into a large epitaxy region. In accordance with some embodiments of the present disclosure, the portions of epitaxy regions 452 grown from neighboring recesses 448 are not merged. This is achieved by making height H4 (FIG. 9B) of fin spacers 446 to be greater than height H2 of fin spacers 246. To make height H4 to be greater than height H2, the period of time TP4 for etching spacer layer 38 (the step shown in FIG. 8B) may be

selected to be shorter than the period of time TP2 for etching spacer layer 38 (the step shown in FIG. 7B). In accordance with some embodiments of the present disclosure, ratio TP2/TP4 may be greater than about 1.5, and may be in the range between about 1.5 and about 5.0. As a result, height H4/H2 may be greater than about 1.5, and may be in the range between about 1.5 and about 5.0.

In accordance with alternative embodiments, the processes for forming fin spacers 236 and 446 may be adjusted to adjust heights H2 and H4, and to result in one of the following results: the merge occurs for both epitaxy regions 252 and 452 (height H2 and H4 are substantially equal, for example, with difference smaller than about 10 percent), the merge occurs for epitaxy regions 452 but not for epitaxy regions 252 (with height H2>H4), and the merge doesn't occur for either of epitaxy regions 252 and 452.

In accordance with some embodiments, after the epitaxy, an implantation is performed to implant a p-type impurity such as boron or indium into epitaxy regions 252 and 452 to form source/drain regions, which are also referred to using reference numerals 252 and 452. In accordance with alternative embodiments, no implantation of p-type impurity is performed.

An etching step is then performed to remove remaining portions of mask layer 56 from regions 100, 200, 300, and 400, and FIGS. 10A, 10B, and 10C illustrate the structure after mask layer 56 is removed. N-type FinFET 166, p-type FinFET 266, n-type FinFET 366, and p-type FinFETs 466 are thus formed in regions 100, 200, 300, and 400, respectively. In subsequent steps, source/drain silicide regions (not shown) are formed on the top surfaces of source/drain regions 152, 252, 352, and 452. An Inter-Layer Dielectric (ILD, not shown) is formed to cover the illustrated FinFETs, and source/drain contact plugs (not shown) may be formed in the ILD to contact the source/drain silicide regions. Gate contact plugs (not shown) may also be formed to contact the illustrated gate electrodes in gate stacks 28. Also, the illustrated gate stacks 28 may be replaced with replacement gate stacks if gate stacks 28 are dummy gate stacks.

The embodiments of the present disclosure have some advantageous features. By separating the formation of fin spacers in different device regions, the heights of the fin spacers in different device regions may be adjusted separately. This advantageously results in the flexibility in forming merged or un-merged epitaxy source/drain regions. The formation of the fin spacers shares a common deposition process, and the epitaxy for forming source/drain regions for different device regions is also a common process. The manufacture cost is thus reduced.

In accordance with some embodiments of the present disclosure, a method includes forming a first gate stack extending on top surfaces and sidewalls of first semiconductor fins with the first semiconductor fins being parallel to and neighboring each other, forming a second gate stack extending on top surfaces and sidewalls of second semiconductor fins with the second semiconductor fins being parallel to and neighboring each other, and forming a dielectric layer. The dielectric layer includes a first portion extending on the first gate stack and the first semiconductor fins, and a second portion extending on the second gate stack and the second semiconductor fins. In a first etching process, the first portion of the dielectric layer is etched to form first fin spacers on sidewalls of the first semiconductor fins. The first fin spacers have a first height. In a second etching process, the second portion of the dielectric layer is etched to form second fin spacers on sidewalls of the second semiconductor

fins. The second fin spacers have a second height greater than the first height. The first semiconductor fins are recessed to form first recesses between the first fin spacers. The second semiconductor fins are recessed to form second recesses between the second fin spacers. The method further includes simultaneously growing first epitaxy semiconductor regions from the first recesses and second epitaxy semiconductor regions from the second recesses. The first epitaxy semiconductor regions grown from neighboring ones of the first recesses merge with each other. The second epitaxy semiconductor regions grown from neighboring ones of the second recesses are separate from each other.

In accordance with some embodiments of the present disclosure, a method includes etching a first semiconductor fin and a second semiconductor fin to form first recesses. The first and the second semiconductor fins have a first distance. A third semiconductor fin and a fourth semiconductor fin are etched to form second recesses. The third and the fourth semiconductor fins have a second distance equal to or smaller than the first distance. An epitaxy is performed to simultaneously grow first epitaxy semiconductor regions from the first recesses and second epitaxy semiconductor regions from the second recesses. The first epitaxy semiconductor regions are merged with each other, and the second epitaxy semiconductor regions are separated from each other.

In accordance with some embodiments of the present disclosure, a method includes, in a common deposition process, forming a dielectric layer including a first portion on top surfaces and sidewalls of first semiconductor fins and a second portion on top surfaces and sidewalls of second semiconductor fins. In separate etching processes, the first portion of the dielectric layer and the second portion of the dielectric layer are etched to form first fin spacers and second fin spacers, respectively. The first fin spacers have a first height, and the second fin spacers have a second height greater than the first height. The first semiconductor fins are etched to form first recesses between the first fin spacers. The second semiconductor fins are etched to form second recesses between the second fin spacers. In a common epitaxy process, first epitaxy semiconductor regions are grown from the first recesses, and second epitaxy semiconductor regions are grown from the second recesses. The first epitaxy semiconductor regions merge with each other, and the second epitaxy semiconductor regions are discrete from each other.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method comprising:

depositing a spacer layer comprising first portions on first semiconductor fins, and second portions on second semiconductor fins;
forming first fin spacers on opposing sides of the first semiconductor fins;

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forming second fin spacers on opposing sides of the second semiconductor fins, wherein the second fin spacers are taller than the first fin spacers; and performing an epitaxy process to simultaneously grow first epitaxy semiconductor regions based on the first semiconductor fins and second epitaxy semiconductor regions based on the second semiconductor fins, wherein when the first epitaxy semiconductor regions grown from neighboring ones of the first semiconductor fins start to merge with each other, the second epitaxy semiconductor regions grown from neighboring ones of the second semiconductor fins are separated from each other.

2. The method of claim 1, wherein at a time when the epitaxy process is finished, the second epitaxy semiconductor regions grown from neighboring ones of the second semiconductor fins are separated from each other.

3. The method of claim 1, wherein the forming the first fin spacers comprises performing a first etching process on a first spacer layer, and the forming the second fin spacers comprises performing a second etching process on a second spacer layer.

4. The method of claim 3, wherein the first etching process and the second etching process are separate etching processes.

5. The method of claim 4, wherein the first etching process lasts longer than the second etching process.

6. The method of claim 1, wherein the first fin spacers have a first height, and the second fin spacers have a second height greater than the first height.

7. The method of claim 6, wherein the second height of the second fin spacers is greater than about 1.5 times the first height of the first fin spacers.

8. The method of claim 1, wherein neighboring fins in the first semiconductor fins have a first distance, and neighboring fins in the second semiconductor fins have a second distance equal to the first distance.

9. The method of claim 1, wherein neighboring fins in the first semiconductor fins have a first distance, and neighboring fins in the second semiconductor fins have a second distance smaller than the first distance.

10. The method of claim 1, wherein the first semiconductor fins overlap first semiconductor strips that are in first isolation regions, and the second semiconductor fins overlap second semiconductor strips that are in second isolation regions, and wherein the method further comprises:

etching the first semiconductor fins to form first recesses between the first fin spacers, wherein the first epitaxy semiconductor regions are grown from the first recesses; and

etching the second semiconductor fins to form second recesses between the second fin spacers, wherein the second epitaxy semiconductor regions are grown from the second recesses.

11. A method comprising:

forming first fin spacers on opposing sides of a first semiconductor strip and a second semiconductor strip; forming second fin spacers on opposing sides of a third semiconductor strip and a fourth semiconductor strip, wherein the second fin spacers are taller than the first fin spacers; and

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forming first epitaxy semiconductor regions from the first semiconductor strip and the second semiconductor strip, wherein the first epitaxy semiconductor regions are merged with each other; and

forming second epitaxy semiconductor regions from the third semiconductor strip and the fourth semiconductor strip, and the second epitaxy semiconductor regions are separated from each other when the forming the second epitaxy semiconductor regions is ended.

12. The method of claim 11, wherein the first semiconductor strip and the second semiconductor strip have a first distance from each other, and wherein the third semiconductor strip and the fourth semiconductor strip have a second distance from each other, with the second distance being equal to or smaller than the first distance.

13. The method of claim 12, wherein the second distance is smaller than the first distance.

14. The method of claim 11, wherein the first epitaxy semiconductor regions and the second semiconductor epitaxy regions are comprised in a first Fin Field-Effect Transistor (FinFET) and a second FinFET, respectively.

15. The method of claim 11, wherein the first epitaxy semiconductor regions and the second epitaxy semiconductor regions are epitaxially grown in a same epitaxy process.

16. The method of claim 11 further comprising depositing a dielectric layer on tops and opposing sides of the first, the second, the third, and the fourth semiconductor strips, wherein: the forming the first fin spacers and the forming the second fin spacers comprise a first etching process for forming the first fin spacers, and a second etching process for forming the second fin spacers.

17. The method of claim 16, wherein the first etching process and the second etching process are performed at different times.

18. The method of claim 16, wherein the first fin spacers and the second fin spacers are formed in separate lithography processes using separate photoresists.

19. A method comprising:

forming first fin spacers on sidewalls of first semiconductor fins;

forming second fin spacers on sidewalls of second semiconductor fins, wherein a first spacing between two immediate neighboring ones of the first semiconductor fins is greater than a second spacing between two immediate neighboring ones of the second semiconductor fins, and wherein the first fin spacers have a first height, and the second fin spacers have a second height different from the first height;

recessing the first semiconductor fins;

recessing the second semiconductor fins; and

in a common epitaxy process, growing first epitaxy semiconductor regions from the recessed first semiconductor fins and second epitaxy semiconductor regions from the recessed second semiconductor fins.

20. The method of claim 19, wherein at a time the common epitaxy process is ended, the first epitaxy semiconductor regions merge with each other, and the second epitaxy semiconductor regions are separate from each other.

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